



ANSWERS TO THE PROBLEMS

“What speaks to the soul escapes our measurements.”

Alexander von Humboldt

Cuba 2026 National Chemistry Competition**Answers to the problems****Authors**

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Logo

National Chemistry Pre-selection

“The year 2025 was historic for Cuban students participating in the chemistry olympiads. For the first time, it was possible to participate in all international events with full Cuban teams. The results were also remarkable. We are very happy with what we have achieved.

We want this exam to be a tribute to the main protagonists: Juan Carlos, Roy Roger, Marco Antonio, Ernesto Alejandro, Leticia María, Bárbaro Eriel, Osmani Lázaro, Melissa María, Cynthia María, Pedro Alexis, Nathan David, Alejandro Murad, Jennifer García, Carlos Manuel and Brayan.

We would also like to thank mentors, family members, managers, former Olympians, friends and colleagues from all over the world for their help in the training of our teams.

The authors

January 2026

Free Distribution

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Distribution of the questions

	First day of exam									
	P01 (15%)	P02 (18%)	P03 (10%)	P04 (15%)	P05 (10%)	P06 (10%)	P07 (12%)	P08 (15%)	P09 (8%)	P10 (15%)
10th	X	-	-	X	X	X	-	-	-	-
11th	-	X	-	-	X	-	X	X	-	-
12th	-	-	X	-	-	-	X	X	X	X
	Second day of the exam									
	P11 (15%)	P12 (15%)	P13 (20%)	P14 (23%)	P15 (12%)	P16 (20%)	P17 (20%)	P18 (20%)		
10th	X	X	X	-	-	-	-	-		
11th	-	-	-	X	X	X	-	-		
12th	-	-	X	-	-	-	X	X		

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gas' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ $= 0,08205 \text{ atm} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero degrees Celsius	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs Energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First-order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Drop in vapor pressure	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volume of a sphere	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Area of a sphere	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Area of a circle	$A_{\text{sp}} = \pi \cdot r^2$	Speed of light	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Long. of a circunf.	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Force of gravity	$F_g = m \cdot g$	Gravity on Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$			
$1 \text{ bar} = 100 \text{ kPa}$			
$1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$			
$1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$			
$1 \text{ dm}^3 = 1 \text{ L}$			
$1 \text{ mL} = 1 \text{ cm}^3$			
$1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]
*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97				
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]				

Specific Instructions for the Chemistry Competition

Dear Student,

- This exam was prepared by a Scientific Committee composed of former Olympians.
- **Fill in your personal details.** Verify your identity with your ID card.
- The provincial board of education is responsible for ensuring that the exam is printed to a sufficient quality, allowing students to read the instructions, questions, figures, data, graphs, and tables. **No questions will be canceled due to bad impressions.**
- Write all your data clearly in the indicated place (names, surnames, school, municipality, province).
- **Do not write anything in the CODE field;** it is for the exclusive use of the person responsible for coding.
- **Verify that your exam is complete.** These are the tasks for the first day of the contest (P1-10).
 - 10th-graders must answer questions 1, 4, 5, and 6.
 - 11th-graders must answer questions 2, 5, 7 and 8.
 - 12th-graders must answer questions 3, 7, 8, 9, and 10.
- **Verify that your exam is complete.** These are the tasks for the second day of the contests (P11-18).
 - 10th-graders must answer questions 11, 12, and 13.
 - 11th-graders must answer questions 14, 15 and 16.
 - 12th-graders must answer questions 13, 17, and 18.
- All necessary constants, conversions, equations, and periodic table are included on the first page of the exam. This page will be deleted during the coding process.
- At the beginning of each question, it is indicated which grade(s) it is aimed at, its value (of the total of 100 points), as well as the value of each item (in marks).
 - You can use a scientific calculator and a ruler.
 - Follow the rules of approximate calculation.
 - Attention! You have limited time, and the exam has many sections.
 - You do not have a space limit to answer, BUT answer in an orderly manner, with appropriately sized and legible font, to facilitate the grading process.
- It is necessary to state all auxiliary calculations **whenever necessary** to justify your reasoning and answers properly. **Answers that are not understood will not be graded.**
- List the additional answer sheets at the bottom of the pages, on each page, and indicate the total number of pages, for example, “1 out of 7”, “2 out of 7”, and so on. **We hope that the exam taker (official) will collaborate with the Chemistry students in this distinctive aspect.**
 - If you detect any irregularity, contact the exam taker immediately.
 - If the exam taker interferes with any of the above guidelines, due to ignorance, exercise your right to protest and immediately request the intervention of the competent authority of the Ministry of Education, which is the institution responsible for the logistics of the exam.
 - The final grade is rounded to the units for the competitor’s benefit.

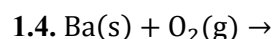
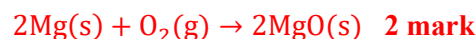
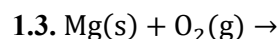
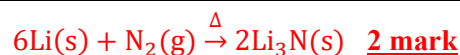
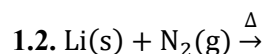
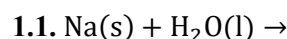
You are a source of pride for Cuban education: Success!

Question 1. Inorganic reactions (15% of total; 10th grade) Day 1

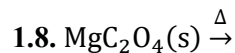
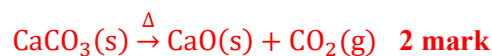
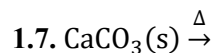
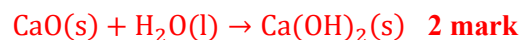
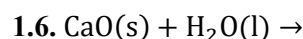
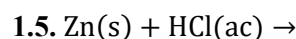
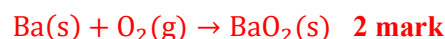
1.1	...	Total	Total
2	Idem	30	15
2	idem	30	15

Chemistry is the science that studies matter, its composition, and its transformations. This discipline did not emerge all at once; it gradually appeared, closely linked to people's daily lives. Since ancient times, people practiced chemistry: although they did not talk about "atoms" or "reactions", they did know how to transform materials. For example: they used the combustion of wood or coal to cook food, heat themselves and in metallurgy; they obtained bread, wine and beer through fermentation; they mixed metals at high temperatures to create alloys such as bronze; they used dyes and pigments to color fabrics and obtain cosmetics; and they used plants, oils and minerals such as perfumes and medicines.

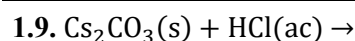
Nowadays, human knowledge has advanced a lot, and it is possible to formulate substances and their transformations. This is one of the most important skills in preparing for the Olympiads. Demonstrate what you have learned in your studies by completing and balancing the following equations.



Hint: The oxidation state of oxygen in the product is 1−.

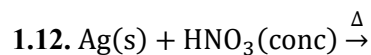
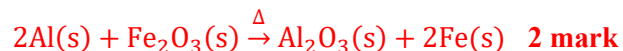
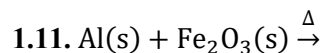
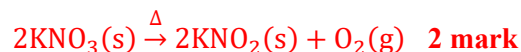


Hint: Two gaseous products are obtained.

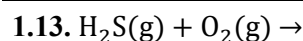
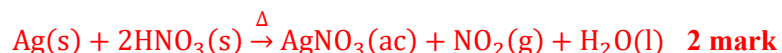




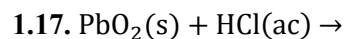
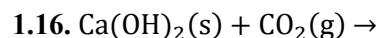
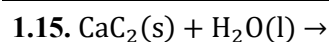
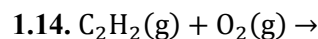
Hint: A new oxysalt is formed.



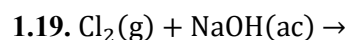
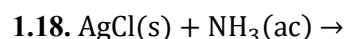
Hint: A brown (reddish-brown) gas is obtained.



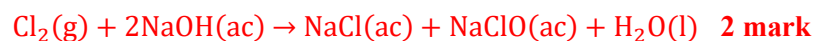
Hint: A gas is obtained in which the sulfur does not reach the maximum oxidation state.



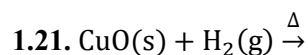
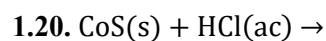
Hint: A green-yellow gas with a characteristic odor is obtained.



Hint: A disproportion reaction occurs.

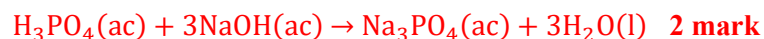
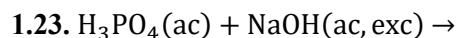
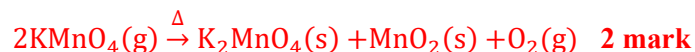


Only 1 mark is awarded if an oxidation state other than +1 (chlorite, chlorate, or perchlorate) is formulated.

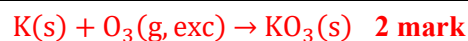
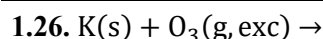
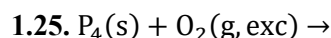
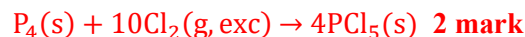
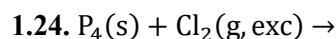




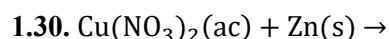
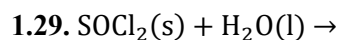
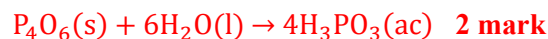
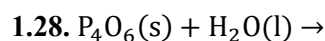
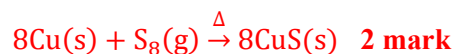
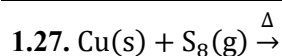
Hint: Products are a gas, a green oxysalt, and a black oxide.



Only 1 mark is awarded if it is formulated as a (di)hydrogen phosphate product.



Only 1 mark is granted in case of formulating peroxide (K_2O_2) or superoxide (KO_2).



Mistakes in oxidation states will not be penalized.

For mistakes in balancing, formula or identifying the correct substance, a penalty of -1 mark is applied. In reactions that yield only one product, if the product is not correctly identified, it is penalized by a -2 mark (no points are awarded).

Question 2. ¡Chemistry is for everyone! (18% of total; only 11th grade) Day 1

2.1	2.2	Total	Total
40	32	72	18
40	32	72	18

One day, the *Captain*, as Naylen is affectionately called, decided to share a coffee with some pre-selection students in the courtyard of her house: Leticia, Melissa, Cynthia, Jennifer, Lauren, Amanda, Gabriela, Britany, Brenda and Diana. She was very upset because she had heard her son Renny say that chemistry was “a man’s thing”. So, together they devised a substance-identification challenge to test the knowledge of the boys’ peers. The next day, they checked the school’s reagent warehouse, where technician Ivis was waiting for them and provided them with some substances. Next, they went to the laboratory, where mentor Agustín had gathered the boys.

“Good morning, teacher,” said the *Captain*. “The ladies on the team bring a challenge to the gentlemen.”

“Whoever does not know how to identify the substances, does not eat the cake that is in the snack that the mothers have prepared,” said Agustín to motivate them. And the parade of substances began.

i. Leticia presented a white solid. “My substance is a ternary salt, whose anion is a halide and can be used as fertilizer, because plants can absorb the cation. It is very soluble in water, resulting in a slightly acidic solution. When heated, it breaks down into two pungent and toxic odorous gases, which quickly recombine when the temperature drops, causing a white fog.” Leticia dissolved a portion in water, and when she added a few drops of AgNO_3 , a white solid precipitated, which dissolved slightly upon adding NaOH to the test tube.

ii. Melissa also brought a flask with a white solid in it. “My substance is a ternary salt; it is used as disinfectant and bleach. It dissolves easily in water and feels soapy to the touch because it is basic. When it reacts with hydrochloric acid, it releases a yellowish-green gas with a very pungent odor. It has an intense yellow coloration in the flame. In solution, it reacts with excess KI , causing an intense yellow-brown coloration.” Melissa added a portion of H_2O_2 , and an intense bubbling of one odorless gas occurred, causing the flame of a match brought close to the lid of the container to intensify.

iii. “Well, I carry a quaternary substance, also white, which is used as a medicine to treat indigestion and in the kitchen as a substitute for baking powder. The aqueous solution of my substance turns bluish green a strip of universal pH indicator paper,” Cynthia said. “My substance dissolves in water, acids, and strong bases. When dissolved in acid, it causes an effervescence capable of extinguishing the flame of a match when it is brought closer to the test tube. In solution, it reacts with calcium chloride, forming a white turbidity.” Cynthia performed the flame test with a combustion spoon, and an intense yellow coloration was observed.

iv. Jennifer brought a flask with a black powder. “My substance is elementary; it occurs in various allotropic forms. The one I have here allows me to write and is also used as a lubricant.” Next, she took portions of the solid and stirred vigorously with water, an acid solution, a basic solution and ethanol; however, no dissolution occurred in any case. When a lit match was brought into contact to a portion of the solid, it burned vigorously. In a test tube under an extraction hood, Jennifer placed the unknown substance and black copper(II) oxide and heated them strongly, obtaining a reddish residue.

v. Lauren showed a colorless crystalline solid. “My substance is a molecular acid, ternary, it exists in the form of dihydrate and is slightly toxic in the body, causing the formation of kidney stones; it is also used as a primary standard in acid-base titrations.” Lauren heated a portion in a test tube, vapors formed that partially condensed on the edge of the tube and extinguished the flame of a match. When an excess of its aqueous solution was added to acidulated KMnO_4 , gradual discoloration was observed that accelerated over time; the discoloration was faster when the experiment was repeated under heating.

vi. Amanda brought a solid yellow color with a pungent smell. “My substance is elemental; it comes in the form of two main allotropes composed of cyclic molecules. It is insoluble in water, but soluble in carbon disulfide. If it burns in the presence of oxygen, it gives off a gas with a pungent odor, which, when bubbling in an orange solution of acidified $\text{K}_2\text{Cr}_2\text{O}_7$, turns it green; moreover, the same gas instantly discolors a violet acidulated solution of KMnO_4 .” Amanda heated a portion of the solid with concentrated NaOH and then carefully added drops of HCl . The rotten-egg-like odor was immediately detected, and turbidity was observed in the test tube.

vii. Gabriela showed a white crystalline solid. “This substance is ternary and is used in pyrotechnics due to its explosive nature. It dissolves in water with difficulty, and the resulting solution is neutral. Its cation causes a violet-lila coloration to the flame. When heated strongly, it decomposes, leaving a white residue and releasing a gas that fuels the combustion of burning splitter.” Gabriela added a portion of concentrated hydrochloric acid to the substance, and a yellowish-green, pungent-smelling gas was released.

viii. The twin Britany brought a crystalline, white solid. “The substance inside the flask is ternary; it is widely used in agriculture as a fertilizer, also as an explosive and in instant cooling packages. Its aqueous solution is slightly acidic. When it is strongly heated, it decomposes, releasing gases that can make a wood chip burn more intensely, producing a characteristic brownish smoke.” To identify the cation and anion present, Britany conducted two separate assays. First, she took a portion of the solid and added it to a test tube containing a concentrated KOH solution. When heated, a colorless gas was released, turning a wet strip of universal pH paper blue. Then, when a few drops of iron(II) sulfate solution were added to the aqueous solution, followed by concentrated sulfuric acid, a lower layer formed, and a brown ring appeared at the interface between the two liquids.

ix. The other twin, Brenda, brought a white solid as well. “This substance is the most abundant on the beach and in deserts. It is very hard, has a high melting point and is used to make glass. It does not dissolve in water, organic substances, acids or weak bases at room temperature, but it does dissolve in HF .” Brenda added a portion to a test tube with concentrated sodium hydroxide and strongly heated the solution. A small portion began to dissolve.

x. “And now it’s my turn!” Diana said, wearing a pair of gloves. She then took what looked like a small, grayish ball from inside a flask and carefully threw it into the bucket of water that Ivis had discreetly placed nearby. And kaboom! The instant the ball entered the water, a yellow flame was observed inside the bucket, and an explosion occurred, spilling all the water and soaking some of the students. Diana quickly pulled out a toy gun loaded with phenolphthalein (HX) solution and began spraying some of the boys, causing their clothes to stain pink.

From such a loud noise, Director Dunierky immediately went up angrily to put things in order. “Gentlemen, what in the world is going on here?” Until he saw half the students soaking wet and the other half’s faces red with laughter. “I knew that this was some invention of the old man... You’d better leave everything very clean, or I’ll send you to field work. And do not make any more noise because in the school there is a visit from the province.” Then he went on his way.

2.1. Help the boys on the team identify the substances that each of the girls had.

Leticia = NH_4Cl **4 mark**

2 mark for identifying the cation as NH_4^+ . 2 mark for identifying the anion as Cl^- .

Melissa = NaClO **4 mark**

2 mark for identifying the cation as Na^+ . 2 mark for identifying the anion as ClO^- ; only 1 mark is awarded if ClO_2^- .

Cynthia = NaHCO_3 **4 mark**

2 mark for identifying the cation as Na^+ . 2 mark for identifying the anion as HCO_3^- .

Jennifer = C **4 mark**

Lauren = $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ **4 mark**

Only 2 mark are awarded if the substance is anhydrous.

Amanda = S_8 **4 mark**

Gabriela = KClO_3 **4 mark**

2 mark for identifying the cation as K^+ . 2 mark for identifying the anion as ClO_3^- ; Only 1 mark is awarded if ClO_4^- , ClO_2^- or ClO^- .

Britany = NH_4NO_3 **4 mark**

2 mark for identifying the cation as NH_4^+ . 2 mark for identifying the anion as NO_3^- .

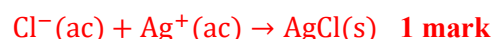
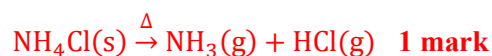
Brenda = SiO_2 **4 mark**

Diana = Na **4 mark**

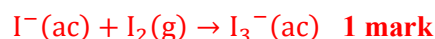
Total = **40 mark**

2.2. Write down the adjusted chemical equations of all the chemical reactions described.

--- Leticia's reactions (NH_4Cl).



--- Melissa's reactions (NaClO).



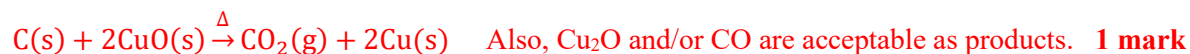
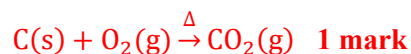
It is considered correct to propose the formation of the species I_3^- directly from I^- .



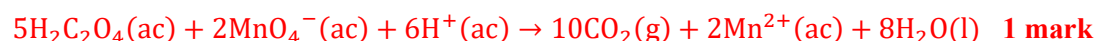
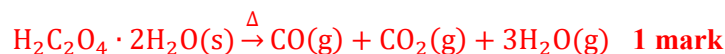
--- Cynthia's reactions (NaHCO_3).



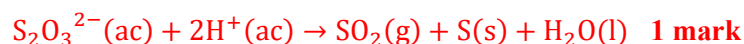
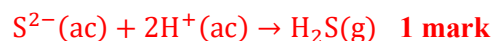
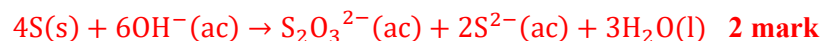
--- Jennifer's reactions (C).



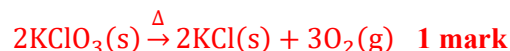
--- Lauren's reactions ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$).



--- Amanda's reactions (S).



--- Gabriela's reactions (KClO₃).



--- Britany's reactions (NH₄NO₃).

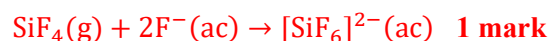


This reaction was already mentioned above and should only be awarded points once.



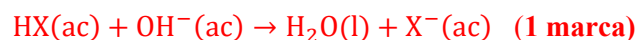
It is acceptable if the complex is formulated with Fe(III) and/or other stoichiometry.

--- Brenda's reactions (SiO₂).



The formation of the species SiF₆²⁻ directly from SiO₂ is considered correct.

--- Diana's reactions (Na).



This last equation is an extra mark in case a mistake was made in any of the previous ones.

Both ionic and molecular equations are accepted. For partially correct equations, 0.5 mark is awarded. You won't be penalized for incorrect aggregation states. S₈ and S are accepted as equivalents. Other logical equations are accepted for the permitted alternative formulations.

Total = 32 mark

Note: Please do not attempt to replicate these experiments at your school or at home.

Question 3. Revenge is colored (10% of total; only 12th grade) **Day 1**

3.1	3.2	Total	Total
12	28	40	10
12	28	40	10

The boys were very thoughtful about what happened in the previous question and agreed that the best way was to answer the girls, “with a little of their own medicine.” Bárbaro, Alejandro *Murad*, Osmany, Brayan, Nathan, Pedro, and the two Carlos came up with a substance-identification question for the girls. Again, the setting was the school’s Chemistry laboratory, but this time, with the principal’s warning not to “cause trouble”. Each student showed a transparent flask containing a colorless aqueous solution of the inorganic substances: AgNO_3 , BaCl_2 , CuSO_4 , Na_2S , VOSO_4 and ZnSO_4 .

Murad and Pedro brought blue solutions. The samples of Bárbaro, Osmany, Brayan and Nathan were colorless. However, Osmany’s had a characteristic odor and Brayan’s, at the end of the demonstration, showed grayish particles when observed against the light. The students then conducted careful assays to provide clues for identification.

Carlos Manuel added a few drops of a sulfuric-acidulated KMnO_4 solution to test tubes containing each of the above substances. The solutions of Bárbaro and Murad were colored purple, with no other apparent change occurring. Brayan’s and Nathan’s solutions were colored purple and became turbid, with the effect much more pronounced in Nathan’s sample. On the contrary, the solution of Osmany caused a yellowish-white turbidity and a very unpleasant odor. Pedro’s solution turned green with the first drops of permanganate, then yellow and finally yellow-violet, when excess reagent was added.

Carlos Daniel used a concentrated ammonia solution for identification. The solution of Bárbaro soon formed a white precipitate that dissolved slightly upon the addition of excess reagent. Murad’s solution produced a sky-blue precipitate, which partially dissolved in excess of the reactant, yielding an intense navy-blue color. The solutions of Osmany and Nathan did not undergo any obvious change. Brayan’s solution immediately formed a white precipitate, which over time turned brown and dissolved in excess of the reactant, resulting in a colorless solution. Pedro’s solution formed a blue-green precipitate that, with excess reagent and prolonged exposure to air, partially dissolved, coloring the solution light yellow. The yellow solution turned orange when a strong acid was added.

3.1. Help the girls on the team identify the substances that each of the boys had.

Bárbaro = ZnSO_4 **2 mark**

Murad = CuSO_4 **2 mark**

Osmany = Na_2S **2 mark**

Brayan = AgNO_3 **2 mark**

Nathan = BaCl_2 **2 mark**

Pedro = VOSO_4 **2 mark**

Total = 12 mark

3.2. Write down the balanced chemical equations of all the chemical reactions described.

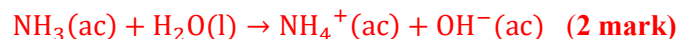
$\text{Ba}^{2+}(\text{ac}) + \text{SO}_4^{2-}(\text{ac}) \rightarrow \text{BaSO}_4(\text{s})$ **2 mark**

$2\text{Ag}^+(\text{ac}) + \text{SO}_4^{2-}(\text{ac}) \rightarrow \text{Ag}_2\text{SO}_4(\text{s})$ **2 mark**

$\text{S}^{2-}(\text{ac}) + 2\text{H}^+(\text{ac}) \rightarrow \text{H}_2\text{S}(\text{g})$ **2 mark**

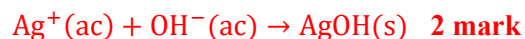
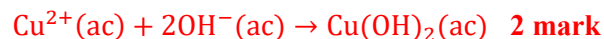
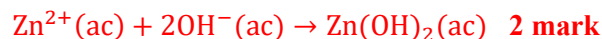
$5\text{S}^{2-}(\text{ac}) + 2\text{MnO}_4^-(\text{ac}) + 16\text{H}^+(\text{ac}) \rightarrow 5\text{S}(\text{s}) + 2\text{Mn}^{2+}(\text{ac}) + 8\text{H}_2\text{O}(\text{l})$ **2 mark**

Only 1 mark is awarded if the sulfide oxidation product is SO_2 or SO_4^{2-} .

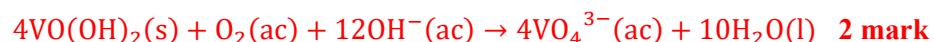
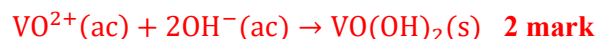


This equation is an extra mark in case a mistake was made in any other.

It is acceptable to formulate the reactions with $\text{NH}_3 + \text{H}_2\text{O}$ (NH_4OH), instead of the active species OH^{-} .



It is also acceptable to join both equations. $2\text{Ag}^{+}(\text{ac}) + 2\text{OH}^{-}(\text{ac}) \rightarrow \text{Ag}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l})$



It is acceptable to propose other vanadium(V) species as metavanadate, VO_3^{-} , or polycondensed ions.

Only 1 mark is awarded if the product formulated for the dissolution of vanadium is $[\text{VO}(\text{NH}_3)_4]^{2+}$, which does not exist.



This last equation is an extra mark in case a mistake was made in any of the others.

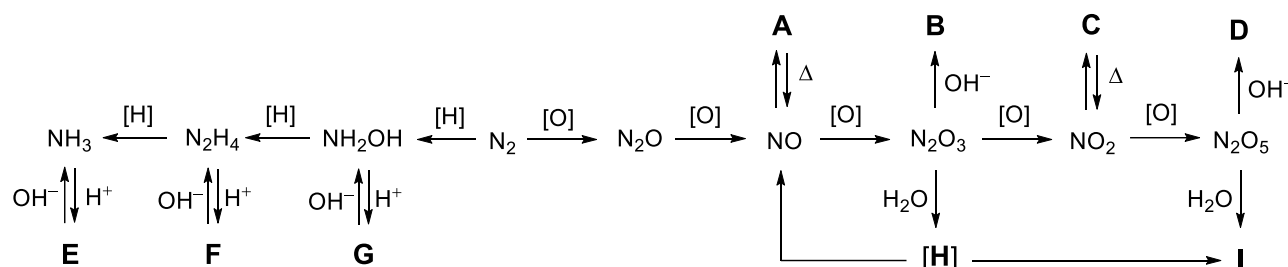
It is acceptable to propose alternative species, such as VO^{3+} or polycondensed ions like $[\text{V}_{10}\text{O}_{28}]^{6-}$.

Total = **28 mark**

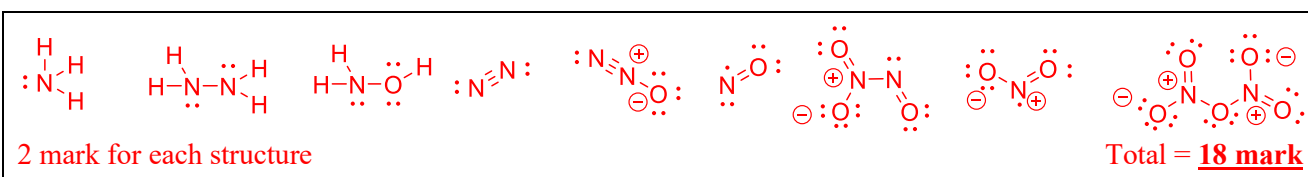
Question 4. Nitrogen (15% of total; only 10th grade) Day 1

4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	Total	Total
18	2	2	2	18	4	6	8	60	15
18	2	2	2	18	4	6	8	60	15

Nitrogen is an element with a very diverse chemistry because it exhibits oxidation states from -3 to $+5$. The diagram below contains some of the most common chemical species.



4.1. Represent the Lewis structures of the nitrogen molecules whose formulas appear in the diagram above.



4.2. Why can ammonia, hydrazine, and hydroxylamine react with H^+ ions to form E, F, and G cations?

These molecules have a pair of free electrons on the N atom, so they are Lewis bases. Total = **2 mark**

4.3. Why can trioxide and pentoxide molecules react with OH^- ions to form B and D anions?

These molecules have a high oxidation state and a large positive partial charge on the nitrogen atom, so that they can accept a pair of free electrons; they are Lewis acids. Total = **2 mark**

4.4. Why can monoxide and dioxide molecules form A and C?

The molecules NO and NO_2 are radicals and tend to dimerize by electron pairing. Total = **2 mark**

4.5. Write the formulas of the substances represented by letters A-I.

A = N_2O_2 B = NO_2^- C = N_2O_4 D = NO_3^- E = NH_4^+ F = N_2H_5^+ G = NH_3OH^+ H = HNO_2 I = HNO_3

2 mark for each formula

Total = **18 mark**

4.6. In concentrated sulfuric medium, the molecules of H and I dehydrate and form the cations J and K, respectively, which are highly reactive species. Write down their formulas.

J = NO^+ K = NO_2^+ Total = **4 mark**

4.7. Explain in detail the different behavior of B and D anions when interacting with transition metals. B easily forms complex ions, preferring to bond at N or O when the oxidation state of the metal is low or high, respectively. In contrast, few complex ions of D are known.

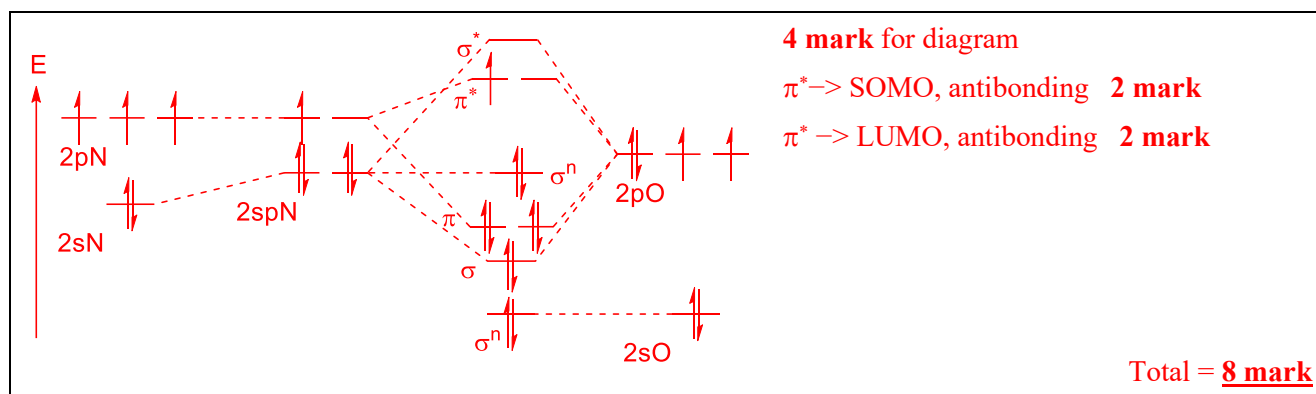
The NO_2^- anion has pairs of unpaired electrons in the nitrogen and oxygen atoms, making it an ambivalent base (ligand). With metals in low oxidation states (soft acids), it prefers to interact with the nitrogen atom (soft center); With metals in high oxidation states (hard acids), the bond preference is with oxygen (hard center). **4 mark**

The NO_3^- anion is stabilized by resonance and is not very reactive. **2 mark**

Total = **6 mark**

4.8. Construct the diagram of molecular orbitals for NO, assuming that nitrogen has sp hybridization. Identify frontier orbitals and rank them according to their type (HOMO, LUMO or SOMO) and contribution to the bond (bonding, anti-bonding or non-bonding).

E (eV): $2s_N = -20,3$ $2p_N = -14,5$ $2s_O = -28,5$ $2p_O = -13,6$

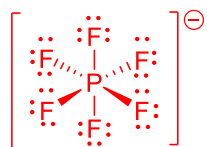


Question 5. Thermochemistry of NaPF₆ (10% of total; only 10th and 11th grades) **Day 1**

5.1	5.2	5.3	5.4	5.5	5.6	Total	Total
4	2	2	2	18	12	40	10
4	2	2	2	18	12	40	10

The hexafluorophosphate anion (PF₆[−]) forms salts that can be used as electrolytes in lithium-ion and sodium batteries. The thermal and chemical stability of these salts is a critical factor in energy storage technologies. Sodium hexafluorophosphate can be obtained by reacting phosphorus pentachloride, sodium chloride and hydrofluoric acid.

5.1. Represent the Lewis structure of the anion. Say the hybridization of phosphorus and the FPF angle.



2 mark hybridization = sp³d² **1 mark** α(FPF) = 90 °, 180 ° **1 mark** Total = **4 mark**

5.2. Pose the equation of the salt formation reaction NaPF₆.



Total = **2 mark**

The lattice energy is a measure of the stability of ionic solids and can be calculated from the Kapustinskii equation below. *U* is the lattice energy in kJ/mol, *r* and *z* are the radii in pm, and the ions' charges, respectively. Σ *v* is the total number of ions in the empirical formula.

$$U \approx -1,202 \cdot 10^5 \cdot \frac{\sum v \cdot |z^+| |z^-|}{r^+ + r^-} \cdot \left(1 - \frac{34,5}{r^+ + r^-}\right)$$

5.3. Propose an expression to estimate the radius of the PF₆[−] ion from the covalent and ionic radii.

$$r(\text{PF}_6^-) \approx r_{\text{cov}}(\text{P}) + r_{\text{cov}}(\text{F}) + r_{\text{ion}}(\text{F}^-)$$

Also $r_{\text{ion}}(\text{P}^{5+}) + 2 \cdot r_{\text{ion}}(\text{F}^-)$ or $r_{\text{cov}}(\text{P}) + r_{\text{cov}}(\text{F}) + r_{\text{vdW}}(\text{F}^-)$ are accepted.

Total = **2 mark**

5.4. Estimate the lattice energy of NaPF₆ if the thermochemical radii of the ions are 102 and 242 pm.

$$U = -1,202 \cdot 10^5 \cdot \frac{2 \cdot 1 \cdot 1}{102 + 242} \cdot \left(1 - \frac{34,5}{102 + 242}\right) = -629 \text{ kJ} \cdot \text{mol}^{-1}$$

Total = **2 marca**

5.5. Use the thermodynamic data **in the table** (energies in kJ/mol, 298 K) and the Born-Haber cycle to calculate the formation enthalpy of NaPF₆. Consider that each P₄ molecule has six P-P bonds.

Sublimation energy of Na(s), ΔH _{sub} (Na)	+108
Primary ionization energy of Na(g), I ₁ (Na)	+496
Sublimation energy of P ₄ (s), ΔH _{sub} (P ₄)	+59
Bond dissociation energy of P-P(g), E(P-P)	+205
Bond dissociation energy of F-F(g), E(F-F)	+155
Energía bond energy of P-F in PF ₆ (g), E(P-F)	+490
Electron affinity energy of the radical PF ₆ (g), ΔH _{AE} (PF ₆)	−770
Lattice energy of NaPF ₆ (s), U(NaPF ₆)	−600
Lattice energy of NaF(s), U(NaF)	−910
Primary electron affinity energy of F(g), A ₁ (F)	−328

By using the Born-Haber cycle for NaPF_6 , we arrive at the following expression.

$$\Delta H_f(\text{NaPF}_6) = \Delta H_{\text{sub}}(\text{Na}) + I_1(\text{Na}) + \frac{1}{4} \cdot \Delta H_{\text{sub}}(\text{P}_4) + \frac{6}{4} \cdot E_{\text{P-P}} + 3 \cdot E_{\text{F-F}} - 6 \cdot E_{\text{P-F}} + \Delta H_{\text{AE}}(\text{PF}_6) + U(\text{NaPF}_6)$$

2 mark for each term (**16 mark** in total for the expression)

$$\Delta H_f = 108 + 496 + \frac{1}{4} \cdot 59 + \frac{6}{4} \cdot 205 + 3 \cdot 155 - 6 \cdot 490 - 770 - 600 = -2919 \frac{\text{kJ}}{\text{mol}} \quad \text{2 mark} \quad \text{Total} = \underline{\text{18 mark}}$$

5.6. Calculate the enthalpy for the thermic decomposition of the complex salt.



It is possible to construct a thermochemical cycle and arrive at the following expression.

$$\Delta H_r = -U(\text{NaPF}_6) - \Delta H_{\text{AE}}(\text{PF}_6) + E_{\text{P-F}} + A_1(\text{F}) + U(\text{NaF})$$

2 mark for each term (**10 mark** in total for the expression)

$$\Delta H_r = +600 + 770 + 490 - 328 - 910 = +622 \text{ kJ} \cdot \text{mol}^{-1} \quad \text{2 mark} \quad \text{Total} = \underline{\text{12 mark}}$$

Alternative answer:

Calculating the formation enthalpy of $\text{NaF}(\text{s})$.

$$\Delta H_f(\text{NaF}) = \Delta H_{\text{sub}}(\text{Na}) + I_1(\text{Na}) + \frac{1}{2} E_{\text{F-F}} + A_1(\text{F}) + U(\text{NaF}) \quad 3 \text{ mark}$$

$$\Delta H_f(\text{NaF}) = 108 + 496 + \frac{1}{2} \cdot 155 - 328 - 910 = -556,5 \frac{\text{kJ}}{\text{mol}} \quad 2 \text{ mark}$$

Calculating the enthalpy of $\text{PF}_5(\text{g})$ formation.

$$\Delta H_f(\text{PF}_5) = \frac{1}{4} \Delta H_{\text{sub}}(\text{P}_4) + \frac{6}{4} E_{\text{P-P}} + \frac{5}{2} E_{\text{F-F}} - 5 E_{\text{P-F}} \quad 3 \text{ mark}$$

$$\Delta H_f(\text{PF}_5) = \frac{1}{4} \cdot 59 + \frac{6}{4} \cdot 205 + \frac{5}{2} \cdot 155 - 5 \cdot 490 = -1740,25 \frac{\text{kJ}}{\text{mol}} \quad 2 \text{ mark}$$

Then, the enthalpy of reaction will be

$$\Delta H_r = \Delta H_f(\text{PF}_5) + \Delta H_f(\text{NaF}) - \Delta H_f(\text{NaPF}_6)$$

$$\Delta H_r = -1740,25 - 556,5 - (-2919) = +622,25 \frac{\text{kJ}}{\text{mol}} \quad 2 \text{ mark}$$

The value of $\Delta H_f(\text{NaPF}_6)$ was already calculated in the previous item.

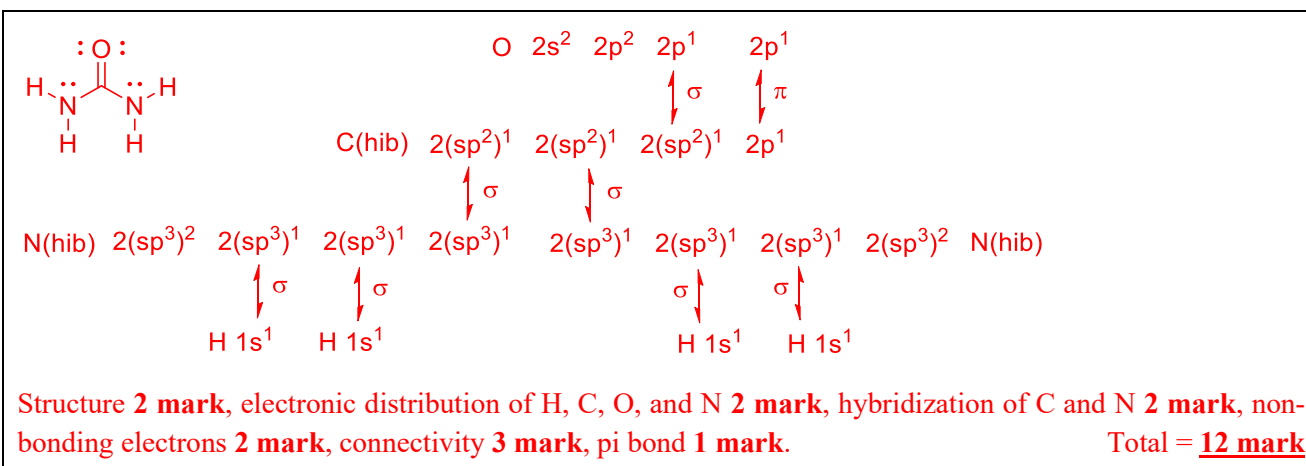
Total = **12 mark**

Question 6. Stoichiometry or chemical calculus (10% of total; only **10th** grade) **Day 1**

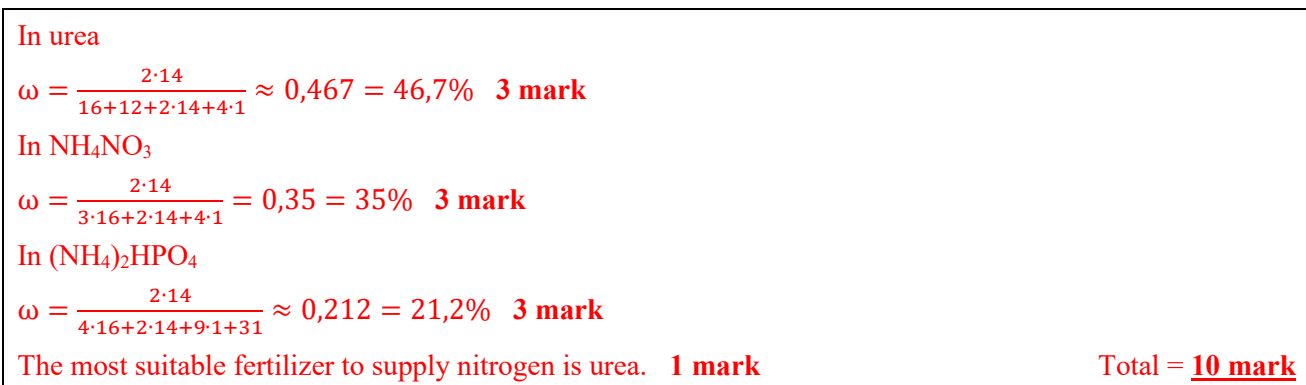
6.1	6.2	6.3	6.4	6.5	Total	Total
12	10	2	12	4	40	10
12	10	2	12	4	40	10

Agriculture demands large quantities of fertilizer to compensate for the nutrient deficit in soil resulting from year-after-year exploitation. The essential nutrients required in greater quantities include the elements N, P and K. Nitrogen is mainly supplied in the form of the organic compound urea, $\text{CO}(\text{NH}_2)_2$ and the inorganic salts NH_4NO_3 and $(\text{NH}_4)_2\text{HPO}_4$. In turn, these fertilizers are obtained from ammonia, an essential product for modern agriculture.

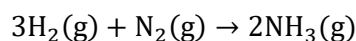
6.1. Represent the Lewis structure of urea. Propose its electronic distribution by the Valence Bond Theory; consider only valence electrons and hybridize C and N atoms.



6.2. Calculate the nitrogen content (in % by mass) in the three fertilizers and say which one is best suited to supply the most nitrogen.

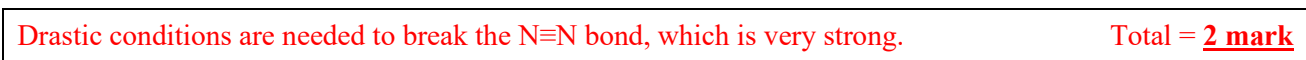


Ammonia is obtained by reacting dinitrogen and dihydrogen under very high temperatures and pressures ($T \sim 720 \text{ K}$, $P \sim 2000 \text{ atm}$) using a heterogeneous iron catalyst.



Typically, it is marketed as a saturated aqueous solution with an ammonia mass content of 30% and a density of 0.90 g/mL .

6.3. Why does ammonia production require high temperatures and the use of a catalyst?



6.4. Suppose you want to get one ton of ammonia from air (21% O₂, 79% N₂, v/v) and hydrogen from electrolysis (decomposition of water by electricity). Calculate the minimum air and water masses required for the process.

It is necessary to calculate the N content in the air and the H content in the water.

$$M(N_2) = 2 \cdot 14 = 28 \text{ g} \cdot \text{mol}^{-1} \quad M(O_2) = 2 \cdot 16 = 32 \text{ g} \cdot \text{mol}^{-1}$$

Assuming 100 moles of air

$$\omega(N, \text{air}) = \frac{79 \cdot 28}{21 \cdot 32 + 79 \cdot 28} = 0,767 \quad \omega(H, \text{water}) = \frac{2 \cdot 1}{2 \cdot 1 + 1 \cdot 16} = 0,111$$

Let's calculate the masses of nitrogen and hydrogen required to produce 1 ton of ammonia.

$$\omega(N, NH_3) = \frac{1 \cdot 14}{1 \cdot 14 + 3 \cdot 1} = 0,824 \rightarrow m(N) = 0,824 \text{ ton} \cdot 10^6 = 8,24 \cdot 10^5 \text{ g}$$

$$m(\text{air}) = \frac{m(N)}{\omega(N, \text{air})} = \frac{8,24 \cdot 10^5}{0,767} = 1,07 \cdot 10^6 \text{ g} = 1070 \text{ kg} = 1,07 \text{ ton} \quad \mathbf{6 \text{ mark}}$$

$$\omega(H, NH_3) = 1 - 0,824 = 0,176 \rightarrow m(H) = 0,176 \text{ ton} \cdot 10^6 = 1,76 \cdot 10^5 \text{ g}$$

$$m(\text{water}) = \frac{m(H)}{\omega(H, \text{water})} = \frac{1,76 \cdot 10^5}{0,111} = 1,59 \cdot 10^6 \text{ g} = 1590 \text{ kg} = 1,59 \text{ ton} \quad \mathbf{6 \text{ mark}}$$

Several possible solutions; points are only awarded for reasoning and final result.

Total = **12 mark**

6.5. Calculate the volume (in L) of an industrial saturated solution containing one ton of ammonia.

$$m(D) = \frac{m(NH_3)}{\omega(NH_3)} = \frac{10^6}{0,30} = 3,33 \cdot 10^6 \text{ g} \rightarrow V(D) = \frac{3,33 \cdot 10^6}{0,90} = 3,7 \cdot 10^6 \text{ mL} \approx 3700 \text{ L} \quad \text{Total} = \mathbf{4 \text{ mark}}$$

Question 7. Back to eleventh Grade (12% of total; only 11th and 12th grades) Day 1

7.1	7.2	7.3	7.4	7.5	7.6	Total	Total
6	8	10	8	8	8	48	12
6	8	10	8	8	8	48	12

7.1. Estimate at what temperature the sea water freezes; given that the mean salinity is 3.5% in mass, the mean density is 1.025 g/mL. Suppose that the only marine salt is NaCl. $K_{cr}(H_2O) = 1.86\text{ }^{\circ}\text{C}\cdot\text{kg}\cdot\text{mol}^{-1}$

It is assumed that all salinity is due to sodium chloride and that dissociation is complete $i = 2$. **1 mark**

$M(\text{NaCl}) = 22,99 + 35,45 = 58,44\text{ g/mol}$ **1 mark**

$$b(\text{NaCl}) = \frac{1000 \cdot \omega\%}{M \cdot (100 - \omega\%)} = \frac{1000 \cdot 3,5}{58,44 \cdot (100 - 3,5)} = 0,621\text{ mol/kg}$$
 2 mark

$$\Delta T = i \cdot b(\text{NaCl}) \cdot K_{cr} = 2 \cdot 0,621 \cdot 1,86 \approx 2,3\text{ }^{\circ}\text{C}$$
 1 mark

Seawater freezes at approximately $-2,3\text{ }^{\circ}\text{C}$. **1 mark**

Total = 6 mark

7.2. In blast furnaces, where metal is produced from the reaction of ore (oxide) with carbon, several equilibria coexist. Three possible reactions of one mole of dioxygen with carbon and carbon monoxide are known. Estimate at what temperature the reduction of the mineral with carbon, forming monoxide, is the most thermodynamically favored reaction among the possible.

Substance	O ₂	C	CO	CO ₂
ΔH_f°	0	0	-111	-394
S°	205	6	198	214

The spontaneity of the reactions is described by $\Delta G^{\circ} = \Delta H^{\circ} - T \cdot \Delta S^{\circ}$.



This reaction has a spontaneity that varies little with temperature.



The spontaneity of this reaction increases with temperature.



The spontaneity of this reaction decreases with temperature.

Comparing reaction 2 with reaction 1, it is necessary to make it more spontaneous.

$$\Delta G^{\circ}(2) = -222 - T \cdot 0,179 < \Delta G^{\circ}(1) = -394 - T \cdot 0,003$$
 1 mark

$$394 - 222 < T \cdot (0,179 - 0,003) \rightarrow T = 977\text{ K}$$
 1 mark

Alternatively, when comparing reaction 2 with reaction 3, for reaction 3 be more spontaneous.

$$\Delta G^{\circ}(2) = -222 - T \cdot 0,179 < \Delta G^{\circ}(3) = -566 + T \cdot 0,173$$

$$566 - 222 < T \cdot (0,179 + 0,173) \rightarrow T = 977\text{ K} = 704\text{ }^{\circ}\text{C}$$

Total = 8 mark

7.3. You want to prepare 1.00 L of aqueous phosphate buffer solution of $\text{pH} = 7.5$ and total molar concentration equal to 0.50 mol/L. An aqueous solution of commercial phosphoric acid ($\sim 15\text{ mol/L}$) and solid sodium hydroxide is available for this purpose. Which are the two species of phosphorus that predominate at this pH value? What amounts do you need to mix of each reagent? For H_3PO_4 we know that $\text{pK}_{a1} = 2.15$, $\text{pK}_{a2} = 7.20$ and $\text{pK}_{a3} = 12.35$.

The amount of phosphoric acid only depends on the total concentration:

$$V(\text{conc}) = \frac{V(\text{dil}) \cdot c(\text{dil})}{c(\text{conc})} = \frac{1000 \cdot 0,50}{15} \approx 33\text{ mL}$$
 2 mark

The desired pH value $\sim \text{pK}_{a2}$, so the predominant species are H_2PO_4^- and HPO_4^{2-} . **2 mark**

$$pK_{a2} = \text{pH} - \log\left(\frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]}\right) \rightarrow 7,20 = 7,50 - \log\left(\frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]}\right)$$

$$[\text{HPO}_4^{2-}] = 2,0 \cdot [\text{H}_2\text{PO}_4^-] \text{ or its equivalent } [\text{H}_2\text{PO}_4^-] = 0,50 \cdot [\text{HPO}_4^{2-}] \quad \mathbf{1 \text{ mark}}$$

$$0,50 = [\text{H}_2\text{PO}_4^-] + [\text{HPO}_4^{2-}] \quad \mathbf{1 \text{ mark}}$$

$$0,50 = [\text{H}_2\text{PO}_4^-] + 2,0 \cdot [\text{H}_2\text{PO}_4^-] \rightarrow [\text{H}_2\text{PO}_4^-] = 0,17 \text{ M} \quad [\text{HPO}_4^{2-}] = 0,33 \text{ M} \quad \mathbf{2 \text{ mark}}$$

With the charge balance, it is possible to calculate the amount of NaOH needed to form the salts.

$$[\text{Na}^+] = [\text{H}_2\text{PO}_4^-] + 2[\text{HPO}_4^{2-}] = 0,17 + 2 \cdot 0,33 = 0,83 \text{ M} \quad \mathbf{1 \text{ mark}}$$

The concentrations of H^+ , OH^- and PO_4^{3-} are not taken into account in the charge balance because they are very small compared to those of the other ions.

$$n(\text{NaOH}) = n([\text{Na}^+]) = 0,83 \cdot 1 = 0,83 \text{ mol} \cdot 40,0 \text{ g/mol} = 33 \text{ g} \quad \mathbf{1 \text{ mark}}$$

Total = **10 mark**

7.4. Most chemical elements form hydroxides and sulfides that are poorly soluble in water. In the 2+ oxidation state, the vast majority of sulfides are more insoluble than hydroxides. In the 3+ and 4+ oxidation states, on the other hand, some sulfides exist only in the solid phase and are rapidly hydrolyzed to form hydroxides in the presence of moisture; for example, this occurs for iron(III) and aluminum. Suppose you have an aqueous solution of MnCl_2 10^{-5} mol/L. Estimate the concentration of sodium sulfide required to initiate precipitation. What product is formed? $K_{ps}(\text{Mn}(\text{OH})_2) = 2,0 \cdot 10^{-13}$, $K_{ps}(\text{MnS}) = 3,0 \cdot 10^{-10}$; for H_2S we know $\text{pK}_{a1} \sim 7,0$ and $\text{pK}_{a2} \sim 13,0$.

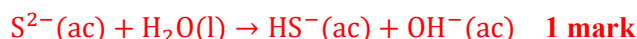
Let's calculate the concentration of sulfur at which MnS begins to precipitate.

$$[\text{S}^{2-}] = \frac{K_{ps}(\text{MnS})}{[\text{Mn}^{2+}]} = \frac{3,0 \cdot 10^{-10}}{10^{-5}} = 3,0 \cdot 10^{-5} \text{ M} \quad \mathbf{1 \text{ mark}}$$

Let's calculate the concentration of hydroxide at which it begins to precipitate $\text{Mn}(\text{OH})_2$.

$$[\text{OH}^-] = \sqrt{\frac{K_{ps}(\text{Mn}(\text{OH})_2)}{[\text{Mn}^{2+}]}} = \sqrt{\frac{2,0 \cdot 10^{-13}}{10^{-5}}} = 1,4 \cdot 10^{-4} \text{ M} \quad \mathbf{1 \text{ mark}}$$

But the concentration of hydroxide is caused by the hydrolysis of sulfide.



$$K_{h2} = \frac{K_w}{K_{a2}} = \frac{1,0 \cdot 10^{-14}}{10^{-13}} = 10^{-1} = \frac{[\text{HS}^-][\text{OH}^-]}{[\text{S}^{2-}]} \approx \frac{[\text{OH}^-]^2}{[\text{S}^{2-}]} \quad \mathbf{1 \text{ mark}}$$

Suppose that $[\text{OH}^-] = 1,4 \cdot 10^{-4} \text{ M}$, then $[\text{S}^{2-}] = 2,0 \cdot 10^{-7} \text{ M}$ and the sulfur has not begun to precipitate. So, the hydroxide precipitates first. **2 mark**

$$c(\text{Na}_2\text{S}) = [\text{S}^{2-}] + [\text{HS}^-] \approx [\text{S}^{2-}] + [\text{OH}^-] = 2,0 \cdot 10^{-7} + 1,4 \cdot 10^{-4} \text{ M} = 1,4 \cdot 10^{-4} \text{ M} \quad \mathbf{2 \text{ mark}}$$

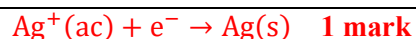
Alternatively, suppose that $[\text{S}^{2-}] = 3,0 \cdot 10^{-5} \text{ M}$, then $[\text{OH}^-] = \sqrt{10^{-1} \cdot 3,0 \cdot 10^{-5}} = 1,7 \cdot 10^{-3} \text{ M}$ and the hydroxide has already begun to precipitate. Total = **8 mark**

7.5. Calculate the potential of the silver-silver chloride reference electrode $\text{Ag}(\text{s})|\text{AgCl}(\text{s})|\text{KCl}(\text{sat.})$ with respect to the normal hydrogen electrode, knowing that $K_{ps}^\circ(\text{AgCl}) = 1,77 \cdot 10^{-10}$, $E^\circ(\text{Ag}^+/\text{Ag}) = 0,7996 \text{ V}$ and that a saturated solution of KCl contains approximately 4.81 mol/L of salt. Note that in such a concentrated solution, the ions interact strongly, and their “effective concentration” or activity is only a factor or fraction of 0.588 (mean ionic activity coefficient) with respect to the molar concentration.

The effective concentration or activity of chlorides shall be:

$$a(\text{Cl}^-) = 0,588 \cdot 4,81 = 2,83 \quad \mathbf{2 \text{ mark}}$$

The reduction semi-equation that best describes this system is



$$E = E^\circ - 0,0591 \cdot \log \left[\frac{1}{[\text{Ag}^+]} \right] \quad \text{1 mark} \quad E = E^\circ - 0,0591 \cdot \log \left[\frac{a(\text{Cl}^-)}{K_{\text{ps}}^\circ} \right] \quad \text{2 mark}$$

$$E = 0,7996 - 0,0591 \cdot \log \left[\frac{2,83}{1,77 \cdot 10^{-10}} \right] = 0,197 \text{ V} \quad \text{2 mark}$$

Total = **8 mark**

7.6. A green dihydrogen production plant uses an electrolyzer consisting of 200 cells. Each cell operates at a current of 5.00 A and requires 1.85 V. The performance or faradaic efficiency of the process is 98.5%, that is, this is the fraction that participates in the desired electrochemical reaction. Calculate the mass of dihydrogen produced each hour and the energy consumption.

The reduction semi-equation that best describes this system is



In each cell, they can theoretically be produced every hour.

$$m(\text{H}_2) = M(\text{H}_2) \cdot n(\text{H}_2) = M(\text{H}_2) \cdot \frac{i \cdot t}{z \cdot F} = 2,02 \cdot \frac{5,00 \cdot 3600}{2 \cdot 96485} = 0,188 \text{ g} \quad \text{3 mark}$$

Considering the efficiency and total number of cells

$$m_{\text{total}}(\text{H}_2) = 0,188 \cdot 0,985 \cdot 200 = 37,0 \text{ g} \quad \text{2 mark}$$

The energy consumption for each hour will be

$$\text{Energy} = Q \cdot \Delta V = i \cdot t \cdot \Delta V = 5,00 \cdot 3600 \cdot 1,85 = 33300 \text{ J} = 9,25 \text{ Wh} \quad \text{2 mark}$$

Total = **8 mark**

Question 8. Ouro Preto (15% of total; 11th and 12th grades) Day 1

8.1	8.2	8.3	8.4	8.5	8.6	8.7	8.8	8.9	8.10	8.11	8.12	8.13	Total	Total
2	8	4	4	2	2	4	3	8	2	9	2	10	60	15
2	8	4	4	2	2	4	3	8	2	9	2	10	60	15

Last year, Cuba participated in the Mendeleev International Olympiad for the second time in Belo Horizonte, Brazil. The “picadillo team” was made up of Juan Carlos (called *Yepe* by everyone), Roy Roger, Marco Antonio, Ernesto Alejandro and Professor Julieta, who is also president of the Cuban Society of Chemistry. The result obtained was very relevant, with a silver medal and three bronze medals. During the event, the team visited the city of Ouro Preto, notorious for gold mining in the late 17th and early 18th centuries. The teacher, who loves chemistry very much, took the opportunity to remind the students of the electrochemical properties of this element.

8.1. Ernesto was struck by the fact that gold is widely used in electronics due to its easy malleability, ductility and excellent electrical conductivity. Why do the triad Cu, Ag and Au have the highest conductivity of all the elements in the periodic table?

These elements have a conduction band formed by the superposition of the orbitals (n-1)d, ns and np, which contains 11 valence electrons. The elements of groups 1-10 contain fewer valence electrons, so they are less conductive. In the elements of the posterior groups, the electrons of the (n-1)d level have very low energy and are strongly retained. Total = **2 mark**

8.2. *Yepe* learned that the city of Ouro Preto received that name because the gold ore found in the area was obscured by a mixture of oxides (approximately 28% and 30%, by mass of oxygen) of a certain transition element. Can you find out which element it is and the formula of its oxide?

Let the formula of the oxides of the unknown element be A_2O_x and A_2O_y .

$$0,28 = \frac{16x}{16x+2A} \rightarrow A = 20,6 \cdot x$$

If $x = 1$, $A = 20,6$. Ne = 20 g/mol, it is ruled out because it is a noble gas.

If $x = 2$, $A = 41,2$. Ca = 40 g/mol; the value is far away, and calcium only forms one oxide.

If $x = 3$, $A = 61,8$. Cu = 63,5 g/mol, but copper only forms oxides in states 1+ and 2+.

If $x = 4$, $A = 82,4$. Kr = 83,8 g/mol, it is ruled out because it is a noble gas.

If $x = 5$, $A = 103$. Rh = 102,9 g/mol. It may be that the most stable oxidation states of Rh are +1 and +3.

$$0,30 = \frac{16y}{16y+2A} \rightarrow B = 18,7 \cdot y \quad \textbf{2 mark}$$

If $y = 1$, $A = 18,7$. F = 19 g/mol; it is discarded because, in nature, it is found in salts.

If $y = 2$, $A = 37,4$. Cl = 35,5 g/mol; it is ruled out because ClO is a gaseous, unstable radical.

If $y = 3$, $A = 56,1$. **Fe = 55,8 g/mol, 2 mark** Fe_2O_3 is a typical iron ore. **2 mark**

If $y = 4$, $A = 74,8$. As = 74,9 g/mol, but arsenic(IV) is not stable.

Checking Fe for mixed oxide Fe_3O_4 , it is found that $\omega = 27,7\%$. **2 mark**

Total = **8 mark**

Julieta really likes jewelry and working with metals. As she is inquisitive, she found out from the organizers what the composition of the gold medals awarded during the event was and the details of the coating process, and she also received one golden medal as a gift.

The medals were made of brass (Cu-Zn alloy), weighed approximately 210 g and had 5.0 μm of gold plating. The gold coating was performed



in a solution of $\text{K}[\text{Au}(\text{CN})_2]$ 0.040 M, at $\text{pH} = 4.0$, using as stabilizer and buffer a solution of 75 g/L of citric acid (H_4Cy , 192 g/mol), at 50°C , at a voltage of 2-4 V, with a current density of 0.50 A/dm^2 and a gold sacrificial anode. The faradaic efficiency of the process is close to 85%.

8.3. Gold(III) cyanide does not exist, but when the two ions come into contact in solution, a pseudohalogen is released, and a precipitate is formed. Formulate the reaction and explain why it happens.



The Au^{3+} is highly polarizing and causes an electronic transfer from the CN^{-} which is highly polarizable. The interaction between Au^{3+} , which is hard acid, and the soft base CN^{-} is not favorable; while the interaction between Au^{+} , which is soft acid, and the soft base CN^{-} is strong. **2 mark** Total = **4 mark**

8.4. $\text{Au}(\text{I})$ is unstable in solution, but can be stabilized by halide ions, cyanide, thiocyanate, and thiosulfate. Why? Calculate the disproportionation constant of $\text{Au}(\text{I})$ in water at 25°C .

Au^{+} is a soft acid and interacts favorably with the halides, cyanide, thiocyanate, and thiosulfate ions, which are soft bases. **2 mark**



$$\Delta E^{\circ} = E^{\circ}(\text{Au}^{+}, \text{Au}) - E^{\circ}(\text{Au}^{3+}, \text{Au}^{+}) = 1,69 - 1,40 = 0,29\text{ V} \quad \text{1 mark}$$

$$K^{\circ} = \exp\left(\frac{2 \cdot 96485 \cdot 0,29}{8,314 \cdot 298}\right) = 6,4 \cdot 10^9 \quad \text{1 mark}$$

Total = **4 mark**

Yepe wanted to know the exact composition of the medals. He determined that the mass of the medal that Julieta was given was 210 g and measured its dimensions, obtaining 40 mm side length and 6.0 mm thick. Using data available online, he was able to determine the composition but did not reveal it to his colleagues.

8.5. Assume for the rest of the question that the medal is shaped like a regular L-sided hexagon, with an area equal to $A = \frac{3\sqrt{3}}{2} \cdot L^2$. Calculate the approximate density of the alloy.

$$V = A \cdot h = \frac{3\sqrt{3}}{2} \cdot L^2 \cdot h = \frac{3\sqrt{3}}{2} \cdot 4^2 \cdot 0,6 = 24,9\text{ cm}^3 \quad \text{1 mark}$$

$$\rho = \frac{m}{V} = \frac{210}{24,9} = 8,43\text{ g/cm}^3 \quad \text{1 mark}$$

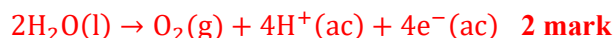
Total = **2 mark**

8.6. The volume obtained by *Yepe* is approximate, because the medal was actually irregularly shaped. Explain a more accurate method of measuring volume in the laboratory.

Dip the medal into a cylinder full of water and calculate the volume change.

Total = **2 mark**

8.7. The efficiency of the gold coating process is often affected because collateral half-reactions occur at the anode. One reaction releases gas and another reaction forms a precipitate that passivates the electrode; thus, requiring higher operating voltage. Formulate the balanced equations of these two reactions.



Total = **4 mark**

Roy then said that his father is a jeweler and that, in his workshop, he uses aqua regia (a mixture of HNO_3 and HCl , in 1:3 proportion) to dissolve certain pieces, because acids alone cannot do so. Julieta explained that this reaction is possible due to the formation of the complex $[\text{AuCl}_4]^{-}$.

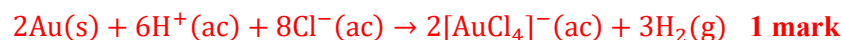
8.8. Help Roy calculate the standard Au^{3+}/Au reduction potential in the absence of chlorides.

$$E^{\circ}(\text{Au}^{3+}, \text{Au}^{+}) = \left(\frac{1,69 + 2 \cdot 1,40}{3}\right) = 1,50\text{ V} \quad \text{3 mark}$$

Total = **3 mark**

8.9. Formulate the possible reactions of gold with H^+ , NO_3^- and O_2 in the air, in the presence of chlorides. Determine which is thermodynamically favored, assuming a medium with $\text{pH} = 0$ and 1.0 M chloride.

In all cases, the formation of the complex must be considered.



$$\Delta E^\circ = E^\circ(\text{H}^+, \text{H}_2) - E^\circ(\text{AuCl}_4^-, \text{Au}) = 0,00 - 1,00 = -1,00 \text{ V} \quad \mathbf{1 \text{ mark}}$$



$$\Delta E^\circ = E^\circ(\text{NO}_3^-, \text{NO}) - E^\circ(\text{AuCl}_4^-, \text{Au}) = 0,96 - 1,00 = -0,04 \text{ V} \quad \mathbf{1 \text{ mark}}$$



$$E'(\text{O}_2, \text{H}_2\text{O}) = E^\circ - \frac{0,0591}{4} \log\left(\frac{1}{p_{\text{r}}(\text{O}_2)}\right) = 1,23 - \frac{0,0591}{4} \log\left(\frac{1}{0,21}\right) = 1,22 \text{ V} \quad \mathbf{1 \text{ mark}}$$

$$\Delta E^\circ = E^\circ(\text{O}_2, \text{H}_2\text{O}) - E^\circ(\text{AuCl}_4^-, \text{Au}) = 1,22 - 1,00 = 0,22 \text{ V} \quad \mathbf{1 \text{ mark}}$$

The difference in potential in the latter case is the most positive of all. **1 mark**

Total = **8 mark**

8.10. In practice, the reaction you found above is not the one that happens. Propose an explanation.

The reaction occurs very slowly because the amount of dioxygen dissolved in the water is small and depends on the rate of diffusion of the gas into the liquid. **2 mark**

Total = **2 mark**

8.11. When Marco calculated the amount of gold (19.3 g/cm^3) in grams consumed in a medal, he was surprised by the result. How long does it take to coat a gold medal?

The total area to be covered is the sum of the two hexagonal faces and the six rectangular side faces

$$A_{\text{total}} = 2A_{\text{cara}} + 6A_{\text{lateral}} = 2 \cdot \frac{3\sqrt{3}}{2} \cdot L^2 + 6 \cdot L \cdot h = 2 \cdot \frac{3\sqrt{3}}{2} \cdot 4^2 + 6 \cdot 4 \cdot 0,6 = 97,5 \text{ cm}^2 \quad \mathbf{2 \text{ mark}}$$

If we disregard the change of area when coating with gold, the volume of the gold coating will be

$$V_{\text{Au}} = A_{\text{total}} \cdot h' = 97,5 \cdot (5 \cdot 10^{-6} \cdot 10^2) = 0,04875 \text{ cm}^3 \quad \mathbf{1 \text{ mark}}$$

$$m(\text{Au}) = \rho_{\text{Au}} \cdot V_{\text{Au}} = 19,3 \cdot 0,04875 = 0,941 \text{ g} = 4,78 \text{ mmol} \quad \mathbf{2 \text{ mark}}$$

Current intensity is calculated as the product of the current density and the deposition area.

$$i = 0,50 \frac{\text{A}}{\text{dm}^2} \cdot 97,5 \text{ cm}^2 \cdot 10^{-2} \frac{\text{dm}^2}{\text{cm}^2} = 0,49 \text{ A} \quad \mathbf{1 \text{ mark}}$$

$$t = \frac{q}{i} = \frac{z \cdot n(\text{Au}) \cdot F}{i} = \frac{1 \cdot 4,78 \cdot 10^{-3} \cdot 96485}{0,49} = 941 \text{ s} \quad \mathbf{2 \text{ mark}} \quad \text{Au(I) is the species that is reduced, thus } z = 1.$$

But considering the efficiency of only 85%

$$t' = \frac{941}{0,85} = 1107 \text{ s} \approx 18,5 \text{ min} \quad \mathbf{1 \text{ mark}}$$

Total = **9 mark**

8.12. At the end of the tour, the teacher, a specialist in organic compounds, asked the students why one of citric acid's constants is extremely small. What would you have answered?

The three acid groups, when deprotonated, form carboxylate anions that are stabilized by resonance. When alcohol is deprotonated, an alkoxide is formed, that is, a non-stabilized anion. **2 mark**

Total = **2 mark**

8.13. A 2.00 g sample of filings from a gold medal was dissolved in hot nitric acid. The concentrated solution was cooled, filtered and neutralized with care to avoid precipitation of the metal hydroxides. Then, it was diluted until it completed a liter of solution in a volumetric flask. A 10 mL aliquot was transferred to an Erlenmeyer bottle, and a small amount of distilled water and 1 g of potassium iodide were added. The resulting suspension was quickly titrated with a 0.0100 M thiosulfate standard, using starch as an indicator.

The volume of titrant required to achieve the first blue coloration was 21.1 mL. What is the total composition (in percent by mass) of the IMChO gold medal?

The chemical reactions that occur during titration are



$$c(\text{Cu}^{2+}) = \frac{c(\text{S}_2\text{O}_3^{2-}) \cdot V(\text{S}_2\text{O}_3^{2-})}{V(\text{Cu}^{2+})} = \frac{0,0100 \cdot 21,1}{10} = 0,0211 \text{ M} \quad \text{2 mark}$$

$$n_{\text{total}}(\text{Cu}) = c(\text{Cu}^{2+}) \cdot V_{\text{D}} = 0,0211 \cdot 1 = 0,0211 \text{ mol} \quad \text{1 mark}$$

$$m_{\text{total}}(\text{Cu}) = n_{\text{total}}(\text{Cu}) \cdot A_{\text{r}}(\text{Cu}) = 0,0211 \cdot 63,5 = 1,34 \text{ g} \quad \text{1 mark} \rightarrow \omega_{\text{Cu}} = 67,0\% \quad \text{2 mark}$$

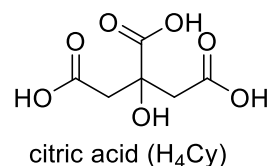
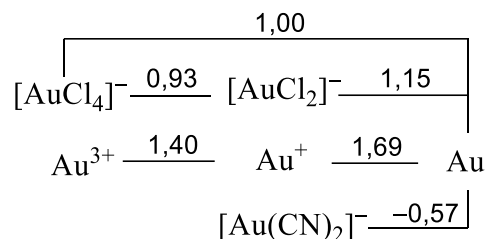
$$m_{\text{medal}}(\text{Cu}) = 0,67 \cdot 210 = 140,7 \text{ g}$$

$$m_{\text{medal}}(\text{Zn}) = 210 - 140,7 - 0,2 = 69,1 \text{ g} \rightarrow \omega_{\text{Zn}} = 32,9\% \quad \text{1 mark}$$

$$\omega_{\text{Au}} = \frac{0,19}{210} \cdot 100 \approx 0,090\% \quad \text{1 mark}$$

It is acceptable to disregard the amount of gold when calculating the percentage of zinc. Total = **10 mark**

Approximate values 67% (Cu), 33% (Zn) and 0,1% (Au) are also correct.



$$K_{\text{ps}}(\text{AuCN}) = 1,0 \cdot 10^{-38} \quad E^{\circ}(\text{V}): \text{O}_2/\text{H}_2\text{O} = 1,23; \text{NO}_3^{-}/\text{NO} = 0,96$$

$$\text{pK}_a(\text{H}_4\text{Cy}) = 3,13; 4,76; 6,40; 14,4 \quad \text{Air: } 79\% \text{ N}_2 \text{ and } 21\% \text{ O}_2 \text{ (v/v), } 1,0 \text{ atm}$$

Note: The author declares that no IMChO medal was damaged during the preparation of this question.

Question 9. Alloys (8% of total; only 12th grade) Day 1

9.1	9.2	9.3	9.4	9.5	9.6	Total	Total
2	6	4	4	12	4	32	8
2	6	4	4	12	4	32	8

Metals can be mixed in different proportions to form alloys, creating new materials. Amalgams are alloys of mercury and are probably among the best known.

9.1. Sodium and zinc amalgam are used in Julia-Lythgoe olefination and Clemmensen reduction, respectively. What is the function of mercury, and why is this specific metal used?

Mercury is a low-active metal and reduces the reactivity of sodium and zinc, allowing them to be handled safely. **1 mark**

Mercury is used to form alloys because it is liquid at room temperature and can diffuse through the solid structure of other metals easily. **1 mark** Total = **2 mark**

The transport of mercury, metallic or in the form of salt, is prohibited on airplanes, because it can cause severe damage to the aluminum structure of the aircraft. The mercury amalgam, Al(Hg), that forms, catalyzes the oxidation of mercury by moisture in the air.

9.2. Why is aluminum inert under normal conditions? Write the equations of the reactions that occur in the presence of mercury, metallic and in the form of salt.

Aluminum has a thin layer of Al(OH)₃ that protects it against oxidation (it is passivated). **1 mark**



Alternative formulations are accepted. Aluminum in the form of amalgam must be specified in at least one of the equations to grant the corresponding mark. Total = **6 mark**

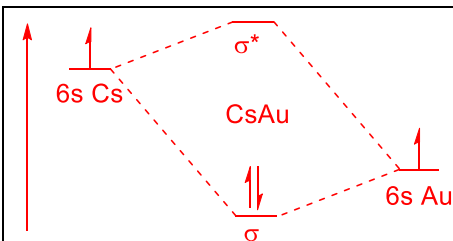
When highly electropositive metals are combined with highly electronegative metals or metalloids, compounds with a high ionic character are created, such as CsAu that melts at 590 °C and the Zintl phases NaTl and CaSi₂, in which the cations are intercalated in a lattice formed by the atoms of the most electronegative element.

9.3. Suggest what kind of three-dimensional structure the species Tl⁻ and Si²⁻ form in the Zintl phases.

The Tl⁻ anion is isoelectronic with the neutral atom of lead, so it forms a three-dimensional diamond-like structure. **2 mark**

The Si²⁻ anion is isoelectronic with the neutral sulfur atom, so it has a preference for forming long chains of simple Si-Si bonds. **2 mark** Total = **4 mark**

9.4. Construct the molecular orbital diagram for the AuCs. Show only the valence orbitals.



Total = **4 mark**

Two metals **A** and **B** from the same group of the periodic table can be mixed in different proportions. When the combination is equimolecular, the mass fraction of **A** is 37%, and a liquid compound is obtained. That liquid is used as a coolant in nuclear reactors for its high thermal conductivity. At low temperatures, it is possible to obtain a solid compound with 54% **A** in mass. Both metals and their compounds explode when they come into contact with water.

9.5. Find the identity of metals **A** and **B**, and the formula for the intermetallic compounds.

Since metals explode in water, they are most likely alkali metals. **1 mark**

An intermetallic compound has the formula **AB** **1 mark** and 37% by mass.

$$0,37 = \frac{A}{A+B} \rightarrow B = 1,7 \cdot A \quad \mathbf{2 \text{ mark}}$$

If $A = \text{Li} = 6,9 \text{ g/mol}$, then $B = 11,7 \text{ g/mol}$, which is not possible (**B** and **C** are in group 1).

If $A = \text{Na} = 23 \text{ g/mol}$ **2 mark**, then $B = 39,1 \text{ g/mol}$, so $B = \text{K}$. **2 mark**

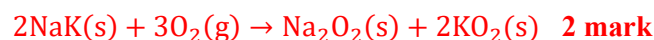
The simplest case of the composition of the other compound is to assume that its formula is Na_xK . **1 mark**

$$0,54 = \frac{x \cdot 23}{x \cdot 23 + 39} \rightarrow x = 2 \quad \mathbf{1 \text{ mark}}$$

The intermetallic compounds are **NaK** and **Na₂K**. **2 mark**

Total = **12 mark**

9.6. Write the balanced equation of the reaction of the equimolecular compound between metals **A** and **B** with excess of: i) water and ii) oxygen.



All marks are awarded for writing reactions with formula **AB**. Half of marks are awarded if reactions are formulated with **Na₂K**. Total = **4 mark**

Question 10. Marginalized elements (15% of total; only 12th grade) **Day 1**To the memory of Tec. Nicasio *Kino* Joaquín Borrego, FQ-UH Organic Department

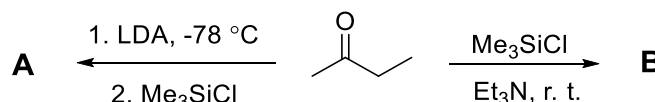
10.1	10.2	10.3	10.4	10.5	10.6	10.7	10.8	Total	Total
6	12	18	2	6	2	4	10	60	15
6	12	18	2	6	2	4	10	60	15

Often, novice students prefer to focus on the organic chemistry of C, H, N, O compounds and are very afraid of molecules with B, P, S, and Si, the four horsemen of the organic chemical apocalypse. This question is meant to unleash their worst fears. But if you have basic knowledge of structure and reactivity, persistence and a little imagination, you will surely be able to answer it.

To answer the following questions, consider the properties of the elements in question.

1. Sulfur compounds have various oxidation states; they can be nucleophilic or electrophilic and stabilize cations and anions. The S atom is soft and can be removed by reduction or oxidation.
2. Boron compounds are electrophilic due to the vacant p-orbital. The B atom has a high affinity for oxygen and can be eliminated by oxidation.
3. Silicon is more electropositive than carbon so that it can stabilize β -carbocations; it is a very hard atom, which has a high affinity for fluorine and oxygen.
4. Phosphorus compounds are restricted to oxidation states 3+ and 5+; they can be nucleophilic or electrophilic. The P atom is soft but has a high affinity for oxygen.

It is possible to control the regioselectivity of enolate formation or its synthetic equivalents by adjusting reaction conditions. Consider the reactions of butanone that form silyl enol ethers.



10.1. Represent the structures of isomeric compounds **A** and **B**. Explain the origin of the selectivity.

A

2 mark

B

2 mark

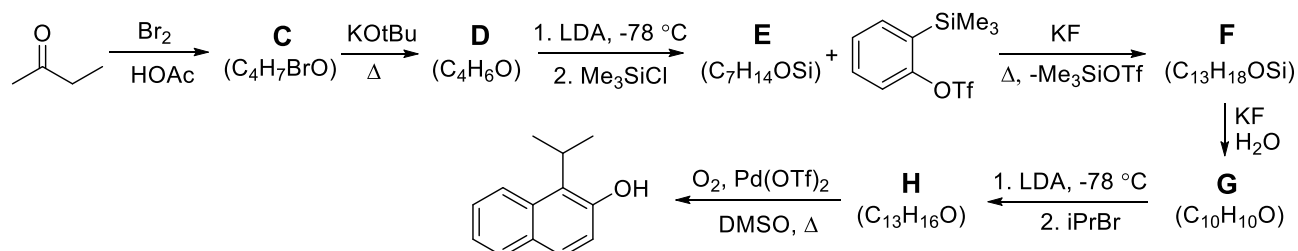
Enol silyl ether **A** is formed under kinetic control at the most acidic and less hindered position of butanone because the temperature is low and the LDA base is very strong and bulky. **1 mark**

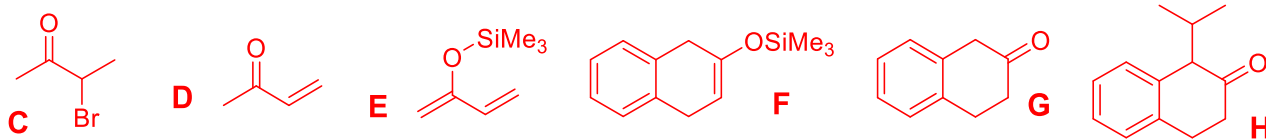
Silyl ether **B** is formed under thermodynamic control in the most stable position because the temperature is high and the Et₃N base is weak and poorly hindered. **1 mark**

Total = 6 mark

Half of the marks are awarded if the A and B isomers are interchanged.

10.2. Consider the following synthesis of a β -naphthol derivative and represent the structures of **C-H**.



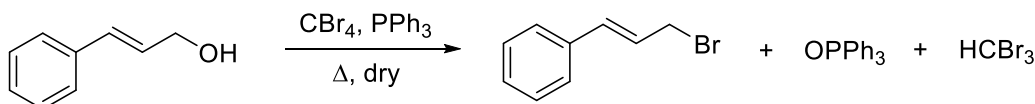


2 mark for each structure

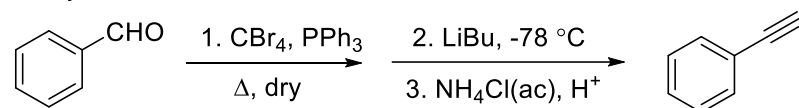
Total = **12 mark**

Phosphorus reagents are frequently used to exchange functional groups in organic reactions. For example, in the reactions of Appel, Corey-Fuch and Mitsunobu.

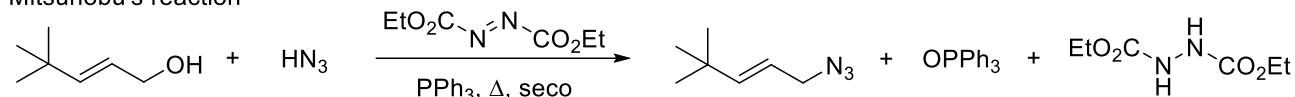
Appel's reaction



Corey-Fuch's reaction

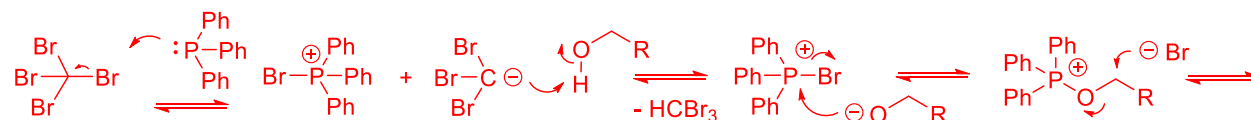


Mitsunobu's reaction

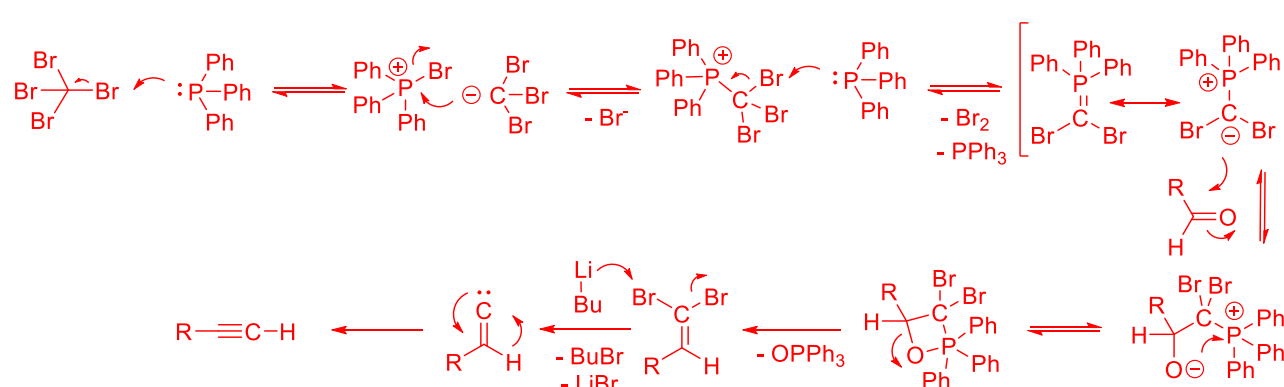


10.3. Propose a mechanism for each of the above reactions. Indicate the movement of the electrons. Why is the use of anhydrous conditions necessary?

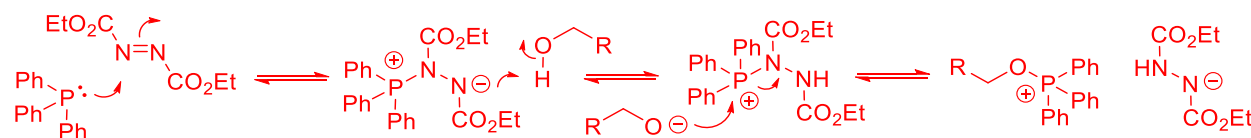
Appel's reaction, 1 mark each step (4 mark)

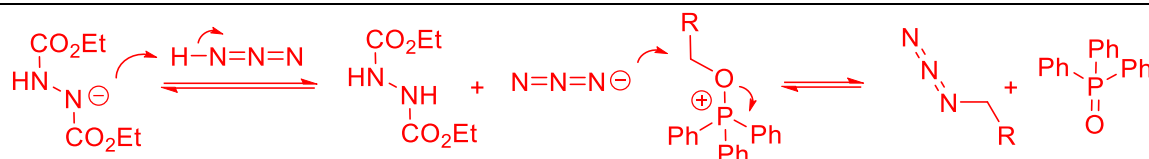


Corey-Fuch's reaction, 1 mark each step (8 mark)



Mitsunobu's reaction, 1 mark each step (5 mark)





Anhydrous conditions prevent hydrolysis of phosphorus intermediates and protonation of anionic intermediates. **1 mark** Total = **18 mark**

10.4. Propose other alternative reagents to transform the allyl alcohol into the corresponding azide.

1. TsCl/MsCl/HX/PX₃/Apple's reaction, 2. NaN₃, aprotic solvent, Δ. **2 mark**

The reaction with HX can lead to transpositions, but this response is accepted.

Total = **2 mark**

The reaction of sodium hydride with boron trichloride in the presence of ether allows the obtaining of two well-known reducing agents **X** and **Y**, depending on the amounts of the reactants. **X** is a molecular gas that rapidly tends to dimerize to **X**₂ and must be stored as dry ether solution, while **Y** is a relatively stable ionic salt that explodes on contact with water.

10.5. Identify substances **X** and **Y** and write the equations for obtaining them.

X = BH₃ **2 mark** **Y** = NaBH₄ **2 mark**

BCl₃(solv) + 3NaH(s) → BH₃(solv) + 3NaCl(s) **1 mark**

BCl₃(solv) + 4NaH(s) → Na[BH₄](s) + 3NaCl(s) or BH₃(solv) + NaH(s) → Na[BH₄](s) **1 mark**

Aggregation status will not be taken into account.

Total = **6 mark**

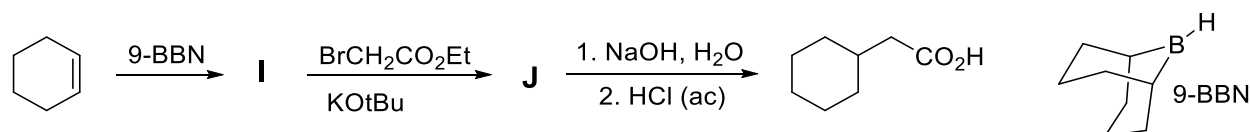
10.6. **Y** is a nucleophilic reductant, which reduces aldehydes, ketones, imines and esters (under heating). In contrast, **X** is an electrophilic reductant, which reduces alkenes, alkynes, carboxylic acids, and amides. Explain the different reactivity between these substances.

Y = NaBH₄ has an excess of negative charge, allowing it to donate electrons to the positively charged carbons of aldehydes, ketones, and imines. **1 mark**

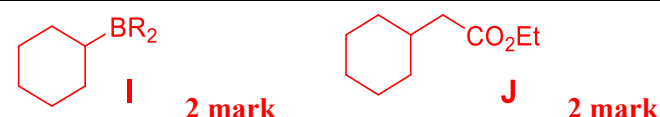
X = BH₃. It features an empty 2p orbital that can accept electrons from nucleophiles such as alkenes, alkynes, and carbonyl groups of acids and amides. **1 mark**

Total = **2 mark**

Boron compounds can be used to synthesize ketones and substituted esters via carbanions, as illustrated in the example.



10.7. Represent the structures of compounds **I** and **J**.

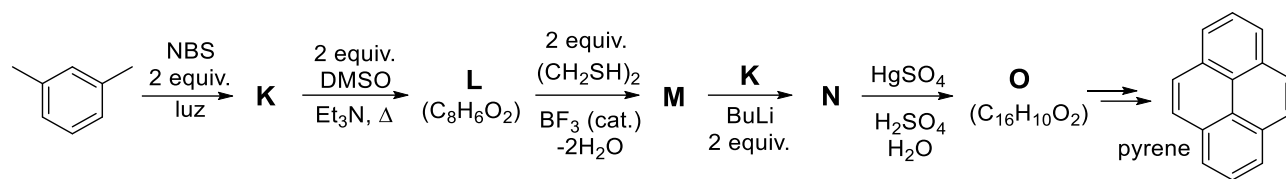


2 mark

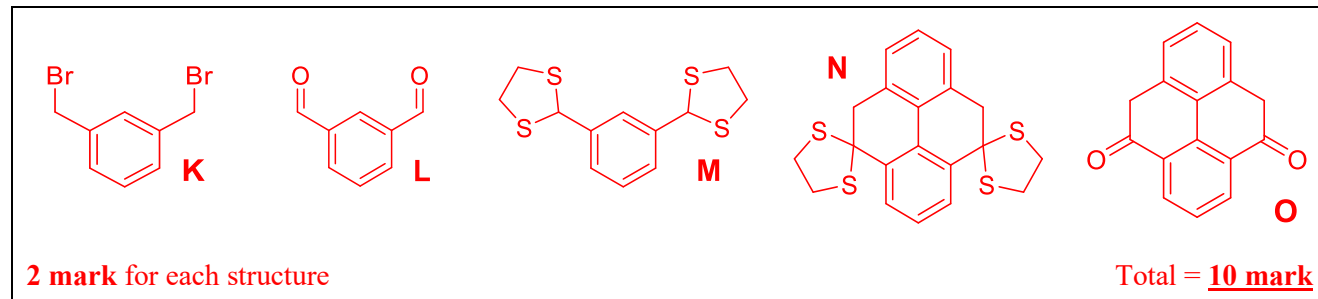
2 mark

Total = **4 mark**

Thiols allow aldehydes (electrophiles) to be masked in the form of dithians (nucleophiles) and to reverse the polarity (umpolung) of chemical reactions. Below you find a possible synthesis of pyrene.



10.8. Represent the structures of **K-O** compounds. NBS: N-bromosuccinimide.



Question 11. Miscellanea are back (15% of total; only 10th grade) Day 2

11.1	11.2	11.3	11.4	11.5	11.6	11.7	11.8	11.9	Total
2	2	2	2	2	2	1	1	1	15
2	2	2	2	2	2	1	1	1	15

The professors of the National Chemistry Pre-selection have decided to test the theoretical knowledge of the new students and have made the following list of experimental facts, which they must explain.

11.1. Luis Guerra: Given the boiling temperatures of hydrogen halides: HF 20 °C, HCl -85 °C, HBr -66 °C and HI -35 °C. Why does HF have an unusually high melting point, whereas the rest of the elements vary in this property in an orderly manner?

HF has hydrogen bonding due to the high difference in electronegativity. **1 point**

Boiling temperatures vary in the order $\text{HCl} < \text{HBr} < \text{HI}$ due to increased dispersion (van der Waals) interactions at larger molecular size/mass. **1 point**

11.2. Rolando: In period 3, the melting temperatures of the elemental substances first increase and then decrease: Na 98 °C, Mg 650 °C, Al 660 °C, Si 1414 °C, P 45 °C, S 119 °C, Cl -102 °C and Ar -189 °C.

Na, Mg, Al and Si are atomic substances, and the particles interact by strong chemical bonds (metallic or covalent); therefore, it takes a lot of energy to melt. **1 point**

P₄, S₈ and Cl₂ are molecular (covalent), and Ar is atomic. In these substances, the interactions between the particles are weak, van der Waals-type and little energy is needed for them to melt. **1 point**

11.3. Landrove: Graphite and graphene conduct the electric current, while diamond is an insulator, even though all these allotropes have the same composition.

In diamond, carbon atoms form localized sigma bonds from sp³ orbitals using all valence electrons, so there is little freedom of movement for the electrons. **1 point**

In graphite and graphene, carbon atoms form strong localized sigma bonds and delocalized pi bonds from sp² orbitals and 2p orbitals, respectively. The electrons that form the pi bond conduct electricity. **1 point**

11.4. Gerardo: The manganese oxides corresponding to oxidation states II and III are basic and reductant, IV is amphoteric and very stable, and VII is very acidic and strongly oxidizing.

As the oxidation number increases, the positive partial charge of the metal increases and the negative partial charge of oxygen decreases. This effect causes an increase in acidity and oxidizing power, while a decrease in basicity and reducing power. **2 point.**

11.5. Rafael: The stability of coordination compounds of the halides depends strongly on the central atom. For example, for iron(III), $[\text{FeF}_6]^{3-} \gg [\text{FeCl}_4]^- > [\text{FeBr}_4]^-$, while for mercury(II): $[\text{HgCl}_4]^{2-} < [\text{HgBr}_4]^{2-} \ll [\text{HgI}_4]^{2-}$. The Fe(III)-I⁻ or Hg(II)-F⁻ complexes are not known.

Pearson's theory states that the interaction between species of equal hardness is the most favorable. The hardness (softness) of the basic halides decreases (increases) in the order $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$. The Fe³⁺ and Hg²⁺ cations are hard and soft acids, respectively, so they interact more strongly with F⁻ and I⁻. **2 point**

11.6. Luis Manuel: Noble gases form very few chemical compounds; however, oxides, fluorides and oxyfluorides of Xe are known.

The Xe element has a lower ionization potential and lower electronegativity among the noble gases and can share its electrons and form bonds with the more electronegative elements O and F. **1 point**

The little size of O and F atoms favors the formation of strong covalent bonds. **1 point**

11.7. Julieta: When the salts potassium iodide and lead(II) nitrate, both white, come into contact, no apparent change occurs. However, adding a few drops of water immediately causes an intense yellow coloration.

The yellow color is due to the reaction: $2\text{KI}(\text{ac}) + \text{Pb}(\text{NO}_3)_2(\text{ac}) \rightarrow \text{PbI}_2(\text{s}) + 2\text{KNO}_3(\text{ac})$

For ion exchange to occur, the presence of water is needed. Solid-state ion mobility is very low and does not allow ion exchange to take place at appreciable speeds. In solution, ion mobility increases significantly and this makes it possible to form lead iodide that precipitates because it is very insoluble. **1 point**

11.8. Eloy: The chemical elements Ga and Al have similar atomic radii, although they are in the same group 13 of the periodic table.

In period 4, the filling of the 3d orbital in the transition elements begins, which causes a cumulative increase in the effective nuclear charge and, as a consequence, a contraction in the radius of the later elements that compensates for the expected increase in radius as one descends the group. **1 point**

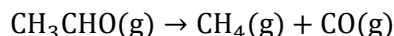
11.9. Rey Michel: The elements boron and silicon are similar in several properties. Both form a covalent atomic crystal lattice; they react with oxygen to form oxides in higher oxidation states and at very high melting temperatures; and they react with halogens to give molecular substances that are strongly hydrolyzed in water, yielding very acidic solutions.

This phenomenon is known as “diagonal periodicity” and is most pronounced between elements in periods 2 and 3. When descending in a group and advancing in a period, the trends of variation in the periodic properties are contrary and partially cancel each other out, allowing the properties of the elements located on a diagonal to be similar. **1 point**

Question 12. Acetaldehyde decomposition (15% of total; only **10th** grade) **Day 2**

12.1	12.2	12.3	12.4	12.5	12.6	12.7	12.8	Total	Total
8	8	12	4	6	6	8	8	60	15
8	8	12	4	6	6	8	8	60	15

The thermal decomposition of acetaldehyde (ethanal) occurs at high temperatures, according to the equation.



The reaction at a certain temperature T was studied, obtaining the following results ($M \equiv \text{mol/L}$).

Experiment	$[\text{CH}_3\text{CHO}]_0$ (M)	Initial rate (M/s)
1	0,10	$1,1 \cdot 10^{-4}$
2	0,20	$4,4 \cdot 10^{-4}$
3	0,30	$9,9 \cdot 10^{-4}$

12.1. Determine the reaction order, the rate law, and the rate constant k_r with the appropriate units.

By simple inspection of the table, it is found that there is a quadratic dependence between the variations in concentration and rate. **4 mark**

The rate law is $v = k_r \cdot [\text{CH}_3\text{CHO}]^2$ **2 mark**

$$k_r = \frac{v}{[\text{CH}_3\text{CHO}]^2} = \frac{1,1 \cdot 10^{-4}}{0,10^2} = 0,011 \text{ M}^{-1}\text{s}^{-1} \quad \text{2 mark}$$

Total = **8 mark**

In other experiments, the dependence of the reaction rate constant, k_r , on temperature, T, in the range 700 to 1000 K was studied. The analysis of the data allowed us to adjust to a straight line when plotting $\ln(k_r)$ versus $(10^3 \cdot \frac{1}{T})$ with slope $-22,5$.

12.2. Calculate the value of the activation energy (in kJ/mol). Assume Arrhenius-like behavior.

$$k_r = A \cdot e^{-\frac{E_A}{RT}} \rightarrow \ln k_r = \ln A - \frac{E_A}{R} \left(\frac{1}{T} \right) \quad \text{2 mark} \rightarrow \ln k_r = \ln A - \frac{E_A}{R \cdot 10^3} \left(\frac{10^3}{T} \right) \quad \text{2 mark}$$

$$-\frac{E_A}{R \cdot 10^3} = -22,5 \rightarrow E_A = 22,5 \cdot 8,314 \cdot 10^3 = 1,87 \cdot 10^5 \frac{\text{J}}{\text{mol}} = 187 \frac{\text{kJ}}{\text{mol}} \quad \text{4 mark} \quad \text{Total = 8 mark}$$

The thermal decomposition of acetaldehyde proceeds via homolytic cleavage of a bond, generating radicals. One of the proposed mechanisms gives dihydrogen and ethane as minority by-products.

	Reaction	E_A (kJ/mol)
Initiation	$\text{CH}_3\text{CHO} \xrightarrow{k_1} \cdot \text{CH}_3 + \cdot \text{CHO}$	358
Propagation	$\text{CH}_3\text{CHO} + \cdot \text{CH}_3 \xrightarrow{k_2} \text{CH}_4 + \text{CH}_3\text{CO} \cdot$	8
	$\text{CH}_3\text{CO} \cdot \xrightarrow{k_3} \cdot \text{CH}_3 + \text{CO}$	59
	$\cdot \text{CHO} \xrightarrow{k_4} \cdot \text{H} + \text{CO}$	65
	$\cdot \text{H} + \text{CH}_3\text{CHO} \xrightarrow{k_5} \text{H}_2 + \text{CH}_3\text{CO} \cdot$	15
Termination	$2 \cdot \text{CH}_3 \xrightarrow{k_6} \text{C}_2\text{H}_6$	0

12.3. Use the steady-state approximation and find mathematical expressions for the variations in the concentrations of the radicals CHO, H, CH₃ and CH₃CO over time (d[X]/dt).

$$\frac{d[\text{CHO}]}{dt} = 0 = k_1[\text{CH}_3\text{CHO}] - k_4[\text{CHO}] \quad \text{2 mark}$$

$$\frac{d[\text{H}]}{dt} = 0 = k_4[\text{CHO}] - k_5[\text{H}][\text{CH}_3\text{CHO}] \quad \text{2 mark}$$

$$\frac{d[\text{CH}_3]}{dt} = 0 = k_1[\text{CH}_3\text{CHO}] + k_3[\text{CH}_3\text{CO}] - k_2[\text{CH}_3\text{CHO}][\text{CH}_3] - 2k_6[\text{CH}_3]^2 \quad \text{5 mark}$$

$$\frac{d[\text{CH}_3\text{CO}]}{dt} = 0 = k_2[\text{CH}_3\text{CHO}][\text{CH}_3] + k_5[\text{H}][\text{CH}_3\text{CHO}] - k_3[\text{CH}_3\text{CO}] \quad \text{3 mark} \quad \text{Total} = \underline{\text{12 mark}}$$

12.4. Why do you think the activation energy of the termination step is zero?

During the reaction, no bonds are broken. **2 mark**

CH₃ radicals are very small, so the steric factor is practically equal to one; that is, the probability that the orientation will be correct during the collision is very high. **2 mark** Total = 4 mark

From the above general mechanism, it is inferred that the decomposition of acetaldehyde occurs by two simpler paths in parallel, with a total of equations.

Mechanism A: $\text{CH}_3\text{CHO}(\text{g}) \rightarrow \text{CH}_4(\text{g}) + \text{CO}(\text{g})$

Mechanism B: $2\text{CH}_3\text{CHO}(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g}) + 2\text{CO}(\text{g}) + \text{H}_2(\text{g})$

12.5. Assign which steps in the general mechanism correspond to each of the simple mechanisms A and B.

Adding the equations of the general mechanism to obtain the total equations, we arrive at

Mechanism A contains steps 2 and 3, **2 mark**

Mechanism B contains the steps 1, 4, 5 and 6 **4 mark** Total = 6 mark

12.6. Taking into account your knowledge of thermodynamics, answer: a) How much is the enthalpy of formation of a simple substance in its standard state? b) What does the standard state symbol (°) mean in enthalpy values? c) What is the meaning of molar caloric capacity?

a) By definition, the enthalpy of formation of a simple substance in the standard state is zero (zero). **2 mark**

b) The symbol (°) means the value was measured at standard pressure (1 bar = 100 kPa ≈ 1 atm). **2 mark**

c) It is the energy needed to increase the temperature of 1 mole by one degree. **2 mark** Total = 6 mark

12.7. Calculate the enthalpy

variations for the total reactions of

Substance	CH ₃ CHO	C ₂ H ₆	CH ₄	CO	H ₂
ΔH_f° [kJ/mol]	-190	-84	-75	-110	?

mechanisms A and B at 298 K from the enthalpy variations in the table.

$$\Delta H_r = \sum(v \cdot \Delta H_f)$$

Mechanism A: $\text{CH}_3\text{CHO}(\text{g}) \rightarrow \text{CH}_4(\text{g}) + \text{CO}(\text{g})$

$$\Delta H_r^{298\text{ K}}(\text{A}) = \Delta H_f(\text{CH}_4) + \Delta H_f(\text{CO}) - \Delta H_f(\text{CH}_3\text{CHO}) \quad \text{2 mark}$$

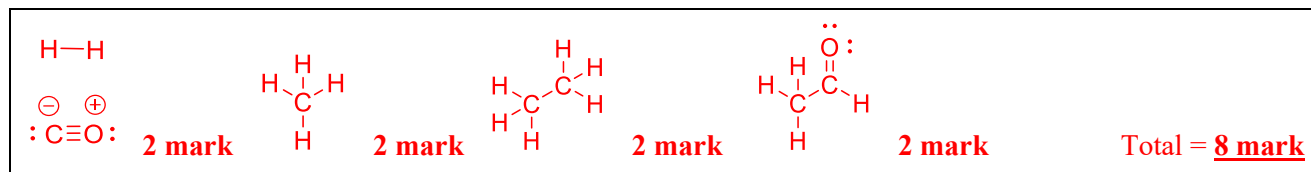
$$\Delta H_r^{298\text{ K}}(\text{A}) = -75 - 110 - (-190) = 5 \text{ kJ} \cdot \text{mol}^{-1} \quad \text{2 mark}$$

Mechanism B: $2\text{CH}_3\text{CHO}(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g}) + 2\text{CO}(\text{g}) + \text{H}_2(\text{g})$

$$\Delta H_r^{298\text{ K}}(\text{B}) = \Delta H_f(\text{C}_2\text{H}_6) + 2 \cdot \Delta H_f(\text{CO}) + \Delta H_f(\text{H}_2) - 2 \cdot \Delta H_f(\text{CH}_3\text{CHO}) \quad \text{2 mark}$$

$$\Delta H_r^{298\text{ K}}(\text{B}) = -84 + 2 \cdot (-110) + 0 - 2 \cdot (-190) = 76 \text{ kJ} \cdot \text{mol}^{-1} \quad \text{2 mark} \quad \text{Total} = \underline{\text{8 mark}}$$

12.8. Represent the Lewis structures of H_2 , CO , CH_4 , C_2H_6 and CH_3CHO .



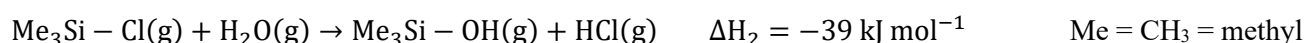
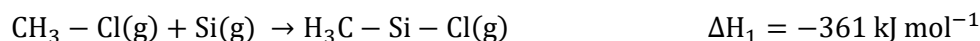
Question 13. Thermochemistry of silicon (20% of total; only 10th and 12th grades) **Day 2**

To the memory of Professor Dr. Jacques Rieumont Briones.

13.1	13.2	13.3	13.4	13.5	13.6	13.7	13.8	13.9	Total	Total
2	18	6	14	12	18	4	2	4	80	20
2	18	6	14	12	18	4	2	4	80	20

Silicon shares characteristics with carbon, such as being in the same family 14, forming long chains (with oxygen in silicones), forming zeolites with catalytic capacity, and having four valence states. This gives it the hypothetical ability to support life forms outside the Earth.

To study the bonds of silicon in its compounds, the heats of the following reactions in the gaseous state at very high temperatures were measured in a pump calorimeter (at constant volume), and using the First Principle of Thermodynamics, the enthalpy changes were calculated.



13.1. What thermodynamic magnitude can be determined directly in a pump calorimeter?

The heat evolved in the calorimeter at constant volume equals the change in internal energy of the chemical reaction. Total = **2 mark**

13.2. Calculate the bond energies E (in kJ/mol) of Si-Cl, Si-C, Si-O and C-Cl, given the values $E(\text{O-H}) = 463$ kJ/mol, $E(\text{Cl-Cl}) = 243$ kJ/mol and $E(\text{Cl-H}) = 431$ kJ/mol.

$$\Delta H_3 = 2 \cdot E(\text{Cl} - \text{Cl}) - 4 \cdot E(\text{Si} - \text{Cl}) \quad \mathbf{2 \text{ mark}}$$

$$-1038 = 2 \cdot 243 - 4 \cdot E(\text{Si} - \text{Cl}) \rightarrow E(\text{Si} - \text{Cl}) = 381 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2 \text{ mark}}$$

$$\Delta H_4 = 2 \cdot E(\text{Cl} - \text{Cl}) - 4 \cdot E(\text{C} - \text{Cl}) \quad \mathbf{2 \text{ mark}}$$

$$-866 = 2 \cdot 243 - 4 \cdot E(\text{C} - \text{Cl}) \rightarrow E(\text{C} - \text{Cl}) = 338 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2 \text{ mark}}$$

$$\Delta H_1 = E(\text{C} - \text{Cl}) - E(\text{Si} - \text{C}) - E(\text{Si} - \text{Cl}) \quad \mathbf{3 \text{ mark}}$$

$$-361 = 338 - E(\text{Si} - \text{C}) - 381 \rightarrow E(\text{Si} - \text{C}) = 318 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2 \text{ mark}}$$

$$\Delta H_2 = E(\text{Si} - \text{Cl}) + E(\text{O} - \text{H}) - E(\text{Si} - \text{O}) - E(\text{Cl} - \text{H}) \quad \mathbf{3 \text{ mark}}$$

$$-39 = 381 + 463 - E(\text{Si} - \text{O}) - 431 \rightarrow E(\text{Si} - \text{O}) = 452 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2 \text{ mark}} \quad \text{Total} = \mathbf{18 \text{ mark}}$$

Silanes ($\text{Si}_n\text{H}_{2n+2}$) are the analogs of alkanes but derived from silicon. Silanes consist of a chain of silicon atoms bonded to hydrogen atoms. The atomization energy (ΔH_{atom}) is the energy needed to break all the bonds of 1 mole of compound, forming only atoms in the gaseous state. For silanes, ΔH_{atom} depends on the number of silicon atoms, as shown below.

Silane	Si_4H_{10}	Si_5H_{12}	Si_6H_{14}	Si_7H_{16}	Si_8H_{18}	Si_9H_{20}
ΔH_{atom} (kJ/mol)	3900	4700	6000	6500	7400	8200

13.3. Find a linear equation that relates ΔH_{atom} to the n number of silicon atoms in the formula of a silane ($\text{Si}_n\text{H}_{2n+2}$) and to the energies of the bonds Si-Si and Si-H.

In the formula $\text{Si}_n\text{H}_{2n+2}$, there are $(2n+2)$ Si-H bonds and $(n-1)$ Si-Si bonds, since this type of bond appears only in disilane. The resulting expression will be

$$\Delta H_{\text{atom}} = (2n + 2) \cdot E(\text{Si} - \text{H}) + (n - 1) \cdot E(\text{Si} - \text{Si}) \quad \mathbf{4 \text{ mark}}$$

$$\Delta H_{\text{atom}} = [2E(\text{Si} - \text{H}) + E(\text{Si} - \text{Si})] \cdot n + [2E(\text{Si} - \text{H}) - E(\text{Si} - \text{Si})] \quad \mathbf{2 \text{ mark}} \quad \text{Total} = \mathbf{6 \text{ mark}}$$

13.4. Calculate $E(\text{Si-Si})$ and $E(\text{Si-H})$ by linear regression or graphically, **as accurately as possible**. Register your calculations and reasoning.

Using **all** the data from the table, the linear regression calculation gives the following results.

intercept $A = 527$ slope $B = 860$ lineality $r = 0,995$ **6 mark** for the regression or the graph

$$2E(\text{Si} - \text{H}) + E(\text{Si} - \text{Si}) = 860 \quad 2E(\text{Si} - \text{H}) - E(\text{Si} - \text{Si}) = 527$$

$$E(\text{Si} - \text{Si}) = 167 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2 \text{ mark}} \quad E(\text{Si} - \text{H}) = 347 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2 \text{ mark}}$$

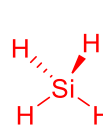
If the values in the table are examined, the atomization energy value for $n = 6$ is a deviation from the general trend. **4 mark** A linear regression without this point provides the most accurate results.

intercept $A = 392$ slope $B = 871$ linearity $r = 0,9998$

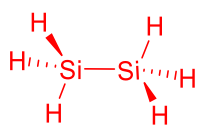
$$2E(\text{Si} - \text{H}) + E(\text{Si} - \text{Si}) = 871 \quad 2E(\text{Si} - \text{H}) - E(\text{Si} - \text{Si}) = 392$$

$$E(\text{Si} - \text{Si}) = \mathbf{240 \text{ kJ} \cdot \text{mol}^{-1}} \quad E(\text{Si} - \text{H}) = \mathbf{316 \text{ kJ} \cdot \text{mol}^{-1}} \quad \text{Total} = \mathbf{14 \text{ mark}}$$

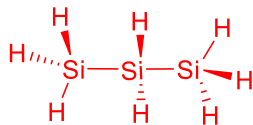
13.5. Represent the structures of silicon hydride ($n = 1$), disilane ($n = 2$), and trisilane ($n = 3$). Calculate your atomization energies.



2 mark



2 mark



2 mark

With the most correct values, you get $\Delta H_{\text{atom}} = 871 \cdot n + 392$

$$n = 1 \rightarrow \Delta H_{\text{atom}} = 1263 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2 \text{ mark}}$$

$$n = 2 \rightarrow \Delta H_{\text{atom}} = 2134 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2 \text{ mark}}$$

$$n = 3 \rightarrow \Delta H_{\text{atom}} = 3005 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2 \text{ mark}}$$

With the least exact values (they are also accepted), you get $\Delta H_{\text{atom}} = 860 \cdot n + 527$

$$n = 1 \rightarrow \Delta H_{\text{atom}} = 1387 \text{ kJ} \cdot \text{mol}^{-1}$$

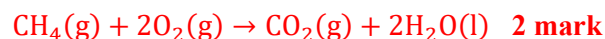
$$n = 2 \rightarrow \Delta H_{\text{atom}} = 2247 \text{ kJ} \cdot \text{mol}^{-1}$$

$$n = 3 \rightarrow \Delta H_{\text{atom}} = 3107 \text{ kJ} \cdot \text{mol}^{-1} \quad \text{Total} = \mathbf{12 \text{ mark}}$$

Methane (CH_4) is stable in air at room temperature and requires an ignition source to initiate combustion. In contrast, its silane analog is pyrophoric, meaning it spontaneously combusts upon contact with air. The structure of the reaction products also differs; while CO_2 is molecular and has two $\text{C}=\text{O}$ bonds, SiO_2 is atomic and has four $\text{Si}-\text{O}$ bonds per structural unit.

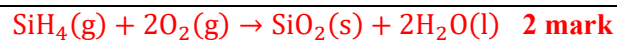
Enlace	C-H	C=O	O-H	O=O	Si-H	Si-O
E (kJ/mol)	413	799	467	498	318	463

13.6. Estimate the enthalpies of combustion for both substances and say which reaction is more favored.



$$\Delta H_{\text{comb}} \approx 4 \cdot E(\text{C} - \text{H}) + 2 \cdot E(\text{O} = \text{O}) - 2 \cdot E(\text{C} = \text{O}) - 4 \cdot E(\text{O} - \text{H}) \quad \mathbf{4 \text{ mark}}$$

$$\Delta H_{\text{comb}} \approx 4 \cdot 413 + 2 \cdot 498 - 2 \cdot 799 - 4 \cdot 467 = -818 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2 \text{ mark}}$$



$$\Delta H_{\text{comb}} \approx 4 \cdot E(\text{Si} - \text{H}) + 2 \cdot E(\text{O} = \text{O}) - 4 \cdot E(\text{Si} - \text{O}) - 4 \cdot E(\text{O} - \text{H}) \quad \text{4 mark}$$

$$\Delta H_{\text{comb}} \approx 4 \cdot 318 + 2 \cdot 498 - 4 \cdot 463 - 4 \cdot 467 = -1452 \text{ kJ} \cdot \text{mol}^{-1} \quad \text{2 mark}$$

The most thermally favored reaction is the combustion of silane. **2 mark** Total = **18 mark**

Ethane ($\text{CH}_3\text{-CH}_3$) is a very stable molecule, with the C-C bond notoriously inert toward water and dilute bases. However, its disilane analog reacts easily with water or aqueous potassium hydroxide, releasing hydrogen.

13.7. Explain the difference in the reactivity of the two substances.

The C-C bonds in ethane are very strong due to the σ -type overlap between the small 2sp^3 orbitals of carbon atoms. **1 mark**

The Si-Si bonds in disilane are weaker due to the σ -type overlap between the larger 3s orbitals of the silicon atoms. **1 mark**

Silicon atoms possess 3d empty orbitals that can accept free electron pairs of Lewis bases such as water and hydroxide ion. **2 mark** Total = **4 mark**

13.8. Write the balanced chemical equation for the hydrolysis reaction of disilane with hydroxide.



13.9. Trimethylamine $\text{N}(\text{CH}_3)_3$ has a pyramidal geometry on the nitrogen atom. However, the tris(trimethylsilyl)amine $\text{N}(\text{Si}(\text{CH}_3)_3)_3$ is trigonal flat. Explain the origin of the different geometries.

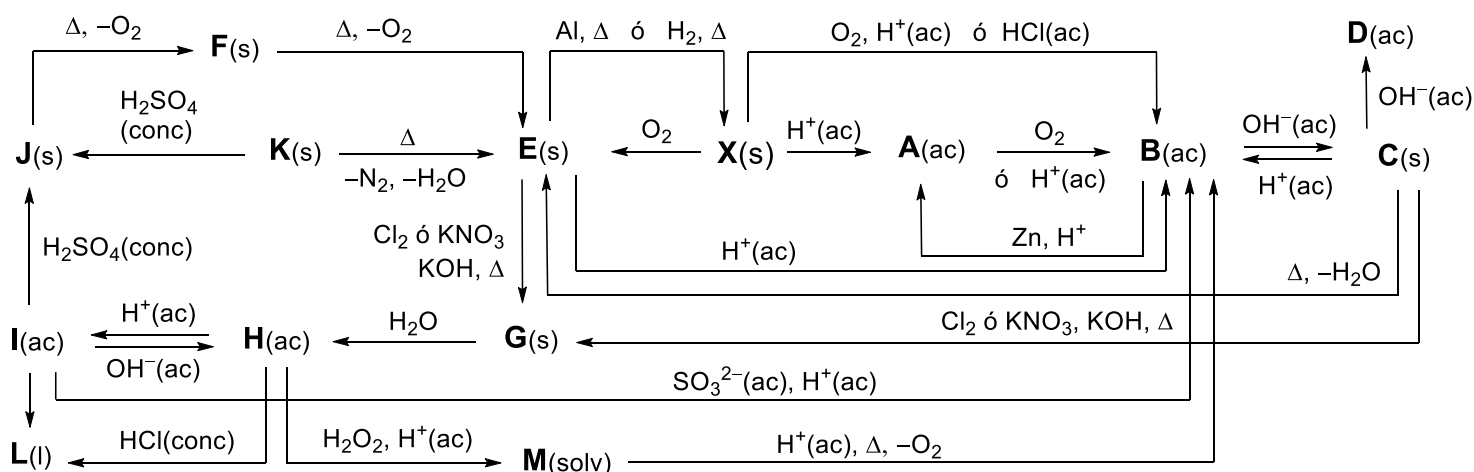
In the $\text{N}(\text{CH}_3)_3$ molecule, nitrogen has sp^3 hybridization, forms three simple bonds of type σ and has a non-bonding lone pair, which determines the pyramidal geometry. **2 mark**

In the molecule $\text{N}(\text{Si}(\text{CH}_3)_3)_3$, nitrogen has sp^2 hybridization, forms three simple bonds of type σ and the extra 2p orbital with two electrons interacts with the empty 3d silicon orbitals, forming weak π bonds, so the bonds have a partial double character, and the geometry is trigonal planar. **2 mark** Total = **4 mark**

Question 14. A transition element (23% of total; only 11th grade) **Day 2**

14.1	14.2	14.3	14.4	14.5	14.6	14.7	14.8	14.9	Total	Total
14	26	10	12	4	6	6	2	12	92	23
14	26	10	12	4	6	6	2	12	92	23

Element **X** is a transition metal of great importance in metallurgy for its hardness and corrosion resistance. The following diagram summarizes its fundamental chemical transformations. All species represented by letters contain the element **X**; species in the aqueous (ac) aggregation state are ions. When the metal is treated with a non-oxidizing acid, it is converted to the reductive cation **A**, which, in the presence of excess acid or air, is rapidly oxidized to the more stable cation **B**. If a solution of **B** becomes basic, it precipitates hydroxide **C**, which is highly insoluble and overreacts in base to give the complex anion **D**. The dehydration of hydroxide forms **E**, which is the most stable oxide of element **X**. Oxide **E** can be obtained by thermal decomposition of the quaternary salt **K**, a very attractive experiment that is known as a “chemical volcano”. When **E** oxide or **C** hydroxide is oxidized (with potassium hypochlorite or nitrate) in basic medium, salt **G** is obtained, containing anion **H**. The anion **I** can be considered as the (dehydrated) dimer of the anion **H**, existing an acid-base equilibrium between the two ions. If dehydration continues, **J** oxide is obtained, which is strongly oxidizing and contains **X** in the maximum oxidation state. The reaction of **H**, **I** or **J** with concentrated hydrochloric acid forms oxohalide **L**, whose molecules have two chlorine atoms. The reaction with hydrogen peroxide in an acidic medium yields **M** oxoperoxide, which is highly unstable and can be isolated only in the presence of organic Lewis bases. It is also known that **F** oxide has magnetic properties and unusual oxidation states of the metal.



14.1. If salt **K** contains 11.1% nitrogen, 3.2% hydrogen and 44.4% oxygen (by mass). Calculate the atomic mass of element **X** and the formula for the salt.

Assuming 100 g of substance **K**

$$n(\text{N}) = \frac{11.1}{14.0} = 0.793 \text{ mol} \quad 1 \text{ mark} \quad n(\text{H}) = \frac{3.2}{1.01} = 3.17 \text{ mol} \quad 1 \text{ mark} \quad n(\text{O}) = \frac{44.4}{16} = 2.78 \text{ mol} \quad 1 \text{ mark}$$

$$n(\text{H}):n(\text{O}):n(\text{N}) = 3.17:2.78:0.793 = 4.00:3.51:1 \approx 8:7:2 \quad 1 \text{ mark}$$

The H: N = 4:1 ratio suggests that **K** is an ammonium salt.

Expressing the formula as the composition of oxides, $\text{K} = 2\text{NH}_3 \cdot \text{H}_2\text{O} \cdot \text{X}_n\text{O}_6$

$$m(\text{X}) = 100 - 11.1 - 3.2 - 44.4 = 41.3 \text{ g} \quad 1 \text{ mark}$$

$$n(X) = \frac{41,3}{A_r(X)} \quad \text{1 mark} \quad \frac{n(X)}{n(N)} = \frac{\frac{41,3}{A_r(X)}}{0,793} = \frac{n}{2} \quad \text{2 mark} \quad \text{Equivalent relationships are accepted.}$$

$$A_r(X) = \frac{41,3 \cdot 2}{0,793 \cdot n} = \frac{104,2}{n} \quad \text{2 mark}$$

For $n=1$, $A_r = 104$ (Rh, Pd?) This result doesn't make chemical sense.

For $n=2$, $A_r = 52,0$ so that **X = Cr (chromium) 2 mark** **K = $(NH_4)_2Cr_2O_7$ 2 mark** **Total = 14 mark**

14.2. State the formulas and color of all the chemical species represented by letters in the diagram.

Hint: **H** and **I** ions in the presence of silver ions form red precipitates.

X = Cr 0 mark **gray 1 mark**

A = Cr^{2+} 1 mark **blue 1 mark**

B = Cr^{3+} 1 mark **green 1 mark**

C = $Cr(OH)_3$ 1 mark **green 1 mark**

D = $[Cr(OH)_4]^-$ 1 mark **bright green 1 mark**

$[Cr(OH)_6]^{3-}$ and CrO_2^- are accepted.

E = Cr_2O_3 1 mark **green 1 mark**

F = CrO_2 1 mark **black 1 mark**

G = K_2CrO_4 1 mark **yellow 1 mark**

H = CrO_4^{2-} 1 mark **yellow 1 mark**

I = $Cr_2O_7^{2-}$ 1 mark **orange 1 mark**

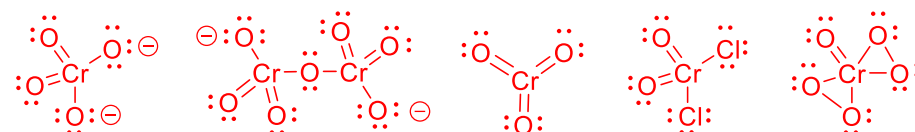
J = CrO_3 1 mark **red 1 mark**

K = $(NH_4)_2Cr_2O_7$ 0 mark **yellow 1 mark**

L = CrO_2Cl_2 1 mark **reddish brown 1 mark**

M = CrO_5 or CrO_5L (L is a Lewis' base, Et_2O , H_2O etc) 1 mark **blue 1 mark** **Total = 26 mark**

14.3. Represent the Lewis structures of species **H**, **I**, **J**, **L** and **M**.



Total = 10 mark

14.4. During the decomposition process $J \rightarrow F$ by heating four oxygenated binary compounds have been isolated, with 54.9, 56.5, 57.5% and 61.9% by mass of **X**, respectively, of which **F** is the one with the lowest oxygen content and is the only stable one. Calculate the formula for the three unstable compounds.

Assuming 100g of each oxide

$$n(Cr) = \frac{54,9}{52,0} = 1,06 \quad \text{1 mark} \quad n(O) = \frac{100-54,9}{16,0} = 2,82 \text{ mol} \quad \text{1 mark}$$

$$n(O):n(Cr) = 2,82:1,06 = 2,66:1 \approx 8:3 \quad \text{1 mark} \quad \text{Cr}_3\text{O}_8 \quad \text{1 mark}$$

$$n(Cr) = \frac{56,5}{52,0} = 1,09 \quad \text{1 mark} \quad n(O) = \frac{100-56,5}{16,0} = 2,72 \text{ mol} \quad \text{1 mark}$$

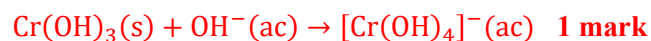
$$n(O):n(Cr) = 2,72:1,09 = 2,50:1 \approx 5:2 \quad \text{1 mark} \quad \text{Cr}_2\text{O}_5 \quad \text{1 mark}$$

$$n(Cr) = \frac{57,5}{52,0} = 1,11 \quad \text{1 mark} \quad n(O) = \frac{100-57,5}{16,0} = 2,66 \text{ mol} \quad \text{1 mark}$$

$$n(O):n(Cr) = 2,66:1,11 = 2,40:1 \approx 12:5 \quad \text{1 mark} \quad \text{Cr}_5\text{O}_{12} \quad \text{1 mark} \quad \text{Total = 12 mark}$$

14.5. What are oxides and hydroxides, such as **E** and **C**, capable of reacting with acids and bases, called? Formulate two equations that illustrate dual behavior with one of these substances as example.

Cr_2O_3 and $Cr(OH)_3$ are amphoteric substances. 2 mark



Molecular equations and those corresponding to Cr_2O_3 .

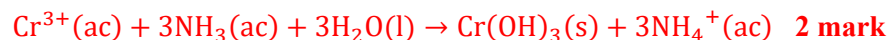
Total = 4 mark

14.6. Tell the type of bond and particle that exists in the oxides **E**, **F** and **J**. Explain this variation.

E is ionic **1 mark** **F** is covalent and atomic **2 mark** **J** is molecular and covalent **2 mark**

As the oxidation number increases, the metal's electronegativity and polarizing power increase; consequently, the Cr-O bond becomes stronger and more covalent. **1 mark** Total = **6 mark**

14.7. What happens when ammonia is added little by little, until an excess is achieved, to a solution of cation **B**? Formulate the corresponding chemical reactions and tell what changes are observed.



A green precipitate is first formed and then dissolved by changing color. **2 mark** Total = **6 mark**

14.8. How can **K** salt be obtained?

By ion exchange and precipitation ("double decomposition") of salts. Also, by adding ammonia to CrO_3 or dichromate salts. Total = **2 mark**

X is found naturally in two minerals, which also contain another metal in the 2+ oxidation state and oxygen. Both minerals can be described as equimolar mixtures (1:1) of two metal oxides. A mineral contains **X** in its most stable oxidation state and 24.9% of an unknown metal **Y**. The other mineral contains **X** in its maximum oxidation state and 64.1% of an unknown metal **Z**.

14.9. Determine which metals **Y** and **Z** are, as well as the empirical formulas for minerals.

In the formula mineral $\text{YO} \cdot \text{Cr}_2\text{O}_3$

$$\frac{A_r(\text{Y})}{M(\text{YO} \cdot \text{Cr}_2\text{O}_3)} = 0,249 = \frac{A_r(\text{Y})}{A_r(\text{Y}) + 16 + 52 \cdot 2 + 16 \cdot 3} \quad \mathbf{2 \text{ mark}} \rightarrow A_r(\text{Y}) = 55,6 \quad \text{Y} = \text{Fe} \quad \mathbf{2 \text{ mark}}$$

In the formula mineral $\text{ZO} \cdot \text{CrO}_3$

$$\frac{A_r(\text{Z})}{M(\text{ZO} \cdot \text{CrO}_3)} = 0,641 = \frac{A_r(\text{Z})}{A_r(\text{Z}) + 16 + 52 + 16 \cdot 3} \quad \mathbf{2 \text{ mark}} \rightarrow A_r(\text{Z}) = 207,1 \quad \text{Z} = \text{Pb} \quad \mathbf{2 \text{ mark}}$$

The formulas of minerals are $\text{FeO} \cdot \text{Cr}_2\text{O}_3 \equiv \text{FeCr}_2\text{O}_4$ **2 mark** and $\text{PbO} \cdot \text{CrO}_3 \equiv \text{PbCrO}_4$ **2 mark**

It is penalized only with -1 mark if $\text{Y} = \text{Mn}$, -1 mark if $\text{Z} = \text{Ag}$. Total = **12 mark**

Question 15. A salty question (12% of total; only 11th grade) Day 2

15.1	15.2	15.3	15.4	15.5	15.6	Total	Total
8	4	6	10	16	4	48	12
8	4	6	10	16	4	48	12

Commercial “low-sodium salts” often consist of mixtures of NaCl (< 30% by mass) and KCl. In these formulations, the element potassium is added to reduce dietary sodium intake. Now imagine that you are a chemical worker in a laboratory, and you must control the quality of a formulation using the following experimental procedure.

1- Dissolve 3,000 g of the commercial salt in distilled water, completing up to 250 mL in a volumetric flask.

2- Take a 50 mL aliquot of the previous solution and add an excess of 0.5 M silver nitrate until quantitative precipitation of the chloride ion is achieved. Filter the precipitate and wash with small portions of ice water. Dry the precipitate by slow heating until constant weight.

15.1. Suppose that during the analysis of a sample **M**, the mass obtained from AgCl precipitate was 1.157 g. Calculate the total chloride content (in % by mass) present in the sample.

$$M(\text{AgCl}) = A_r(\text{Ag}) + A_r(\text{Cl}) = 107,87 + 35,45 = 143,32 \text{ g} \cdot \text{mol}^{-1} \quad \text{1 mark}$$

$$n(\text{AgCl}) = \frac{m(\text{AgCl})}{M(\text{AgCl})} = \frac{1,157}{143,32} = 0,008073 \text{ mol} \quad \text{2 mark}$$

The sample was dissolved in 250 mL, and an aliquot of 50 mL was analyzed, the fifth part. **1 mark**

$$n_{\text{total}}(\text{Cl}) = 0,008073 \cdot 5 = 0,04037 \text{ mol}$$

$$m_{\text{total}}(\text{Cl}) = n_{\text{total}}(\text{Cl}) \cdot A_r(\text{Cl}) = 0,04037 \cdot 35,45 = 1,431 \text{ g} \quad \text{2 mark}$$

$$\omega(\text{Cl}) = \frac{m_{\text{total}}(\text{Cl})}{m(\text{muestra})} = \frac{1,361}{3,000} = 0,4770 \approx 48\% \quad \text{2 mark}$$

Total = **8 mark**

Since the previous method has the disadvantage of losses due to precipitate handling, you now carry out the analysis using reverse titration by the Volhard method.

1- Take a 25 mL aliquot of the sample solution, add a 25 mL aliquot of 0.400 M AgNO₃ standard solution, and add a few drops of acidic iron(III) chloride indicator solution with nitric acid.

2- Titrate with a standard solution of potassium thiocyanate 0.100 M until the appearance of the carmine-red color of the [Fe(SCN)]²⁺ complex.

15.2. Why is it necessary to add acid to the FeCl₃ solution? Write the relevant chemical equation.

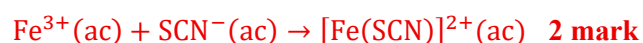
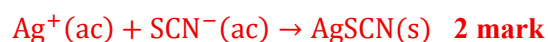
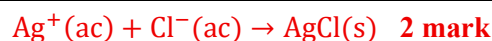
To prevent hydrolysis of the ion, which causes complex ions and insoluble hydroxide.



Other formulations of reactions and compounds are accepted.

Total = **4 mark**

15.3. Write down the equations of all the reactions that occurred during the second determination.



Total = **6 mark**

15.4. Suppose that during the analysis of the same M sample, the consumed volume of thiocyanate titrating reagent was 58.1 mL. Calculate the total chloride content (in % by mass) present in the sample.

$$n_{\text{total}}(\text{AgNO}_3) = c(\text{AgNO}_3) \cdot V(\text{AgNO}_3) = 0,400 \cdot 0,025 = 0,0100 \text{ mol} \quad \text{1 mark}$$

$$n(\text{KSCN}) = c(\text{KSCN}) \cdot V(\text{KSCN}) = 0,100 \cdot 0,0581 = 0,00581 \text{ mol} \quad \text{1 mark}$$

$$n(\text{KSCN}) = n_{\text{exc}}(\text{AgNO}_3) = 0,00581 \text{ mol} \quad \text{1 mark}$$

$$n(\text{AgCl}) = n_{\text{total}}(\text{AgNO}_3) - n_{\text{exc}}(\text{AgNO}_3) = 0,0100 - 0,00581 = 0,00419 \text{ mol} \quad \text{2 mark}$$

The sample was dissolved in 250 mL, and a 25 mL aliquot was analyzed, one-tenth of the total. **1 mark**

$$n_{\text{total}}(\text{Cl}) = 0,00419 \cdot 10 = 0,0419 \text{ mol}$$

$$m_{\text{total}}(\text{Cl}) = n_{\text{total}}(\text{Cl}) \cdot A_r(\text{Cl}) = 0,0419 \cdot 35,45 = 1,485 \text{ g} \quad \text{2 mark}$$

$$\omega(\text{Cl}) = \frac{m_{\text{total}}(\text{Cl})}{m(\text{muestra})} = \frac{1,485}{3,000} = 0,4951 \approx 50\% \quad \text{2 mark}$$

Total = **10 mark**

15.5. Using the data from this second determination, determine the composition (in % by mass) of sample M and say whether it can be classified as “low-sodium”.

$$M(\text{KCl}) = A_r(\text{K}) + A_r(\text{Cl}) = 39,10 + 35,45 = 74,55 \text{ g} \cdot \text{mol}^{-1} \quad \text{1 mark}$$

$$M(\text{NaCl}) = A_r(\text{Na}) + A_r(\text{Cl}) = 22,99 + 35,45 = 58,44 \text{ g} \cdot \text{mol}^{-1} \quad \text{1 mark}$$

$$n_{\text{total}}(\text{Cl}) = n(\text{KCl}) + n(\text{NaCl}) = 0,0419 \quad \text{2 mark}$$

$$\frac{m(\text{KCl})}{74,55} + \frac{m(\text{NaCl})}{58,44} = 0,0419 \quad \text{2 mark} \quad (\text{equation I})$$

$$m(\text{KCl}) + m(\text{NaCl}) = 3,000 \quad \text{2 mark} \quad (\text{equation II})$$

Solving the system of equations yields

$$m(\text{KCl}) = 2,55 \text{ g} \quad \text{2 mark} \quad \rightarrow \quad \omega(\text{KCl}) = \frac{2,55}{3,00} = 85\% \quad \text{1 mark}$$

$$m(\text{NaCl}) = 0,45 \text{ g} \quad \text{2 mark} \quad \rightarrow \quad \omega(\text{NaCl}) = \frac{0,45}{3,00} = 15\% \quad \text{1 mark}$$

The sample can be classified as “low-sodium”. **2 mark**

Total = **16 mark**

15.6. Consider the following data for complex ion formation and solubility product constants. Explain the different affinity of the SCN^- and Cl^- anions to the Ag^+ and Fe^{3+} cations.

$$K_{\text{ps}}(\text{AgCl}) = 1,8 \cdot 10^{-10} \quad K_{\text{ps}}(\text{AgSCN}) = 1,0 \cdot 10^{-12} \quad K_{\text{f}}[\text{Fe}(\text{SCN})]^{2+} = 890 \quad K_{\text{f}}[\text{Fe}(\text{Cl})]^{2+} = 115$$

The anions SCN^- and Cl^- are classified as soft bases (intermediate in the case of chloride). **1 mark**

The Ag^+ cation is a soft acid, and the Fe^{3+} cation is a hard acid. **2 mark**

The more favored interaction will be between acids and bases of equal hardness/softness. **1 mark**

Total = **4 mark**

Question 16. Phosphate determination (20% of total; only **11th** grade) **Day 2**

16.1	16.2	16.3	16.4	16.5	16.6	16.7	16.8	Total	Total
4	12	6	12	10	8	6	22	80	20
4	12	6	12	10	8	6	22	80	20

Breakfast cereals are usually fortified with essential minerals. In this analysis, the goal is to determine the content of inorganic phosphorus (in the form of phosphate) using three different techniques: neutralization volumetry, spectrophotometry and gravimetry.

A 5.0000 g sample of a commercial cereal was pulverized and incinerated. The ashes were dissolved in a few drops of concentrated HCl, and distilled water was added. It was gently heated to remove excess volatile acid. After cooling, the resulting solution is transferred to a 250 mL volumetric flask, and distilled water is added to bring the volume to the mark (**M₀ Stock Solution**). In this step, all the phosphate was transformed into dissolved H₃PO₄.

Part I. Initially, a 25.0 mL aliquot of the **M₀** solution was taken, and 10.0 mL of H₂SO₄ 0.0130 M was added. This mixture was diluted with distilled water to a final volume of 100 mL in a volumetric flask (**M₁ Solution**). An aliquot of 25.0 mL of the new solution was titrated with 0.0151 M NaOH, using thymolphthalein as an indicator (turning interval 9.3-10.5). The average volume of the base to reach the endpoint of the titration was 8.5 mL. It is known that the pH of the final solution of the titration was close to 10. Consider that the only protolytically active component present in the solution obtained from the cereal is phosphate. pK_a(H₃PO₄) = 2,15; 7,20; 12,32 a 25 °C.

16.1. Which of the phosphoric acid species will be present in greater proportion at the beginning and at the end point of the titration?

Beginning = H₃PO₄ **2 mark** End = HPO₄⁻ **2 mark**

Total = **4 mark**

16.2. Calculate the molar concentration of phosphate in solution **M₁**.

It is necessary to know the amount of sulfuric acid present in the 25 mL aliquot of **M₁**.

$$c_{M_1}(H_2SO_4) = \frac{c_{conc}(H_2SO_4) \cdot V_{conc}(H_2SO_4)}{V_{total}(M_1)} = \frac{0,0130 \cdot 10,0}{100} = 0,00130 \text{ M} \quad \mathbf{2 \text{ mark}}$$

$$n_{reac}(H_2SO_4) = c_{M_1}(H_2SO_4) \cdot V_{aliquota}(M_1) = 0,00130 \cdot 25 = 0,0325 \text{ mmol} \quad \mathbf{2 \text{ mark}}$$

Sodium hydroxide reacts with sulfuric and phosphoric acids to form sulfate and hydrogen phosphate.

$$n(NaOH) = c(NaOH) \cdot V(NaOH) = 0,0151 \cdot 8,5 = 0,128 \text{ mmol} \quad \mathbf{2 \text{ mark}}$$

$$n(NaOH) = 2 \cdot n_{reac}(H_2SO_4) + 2 \cdot n(H_3PO_4) \quad \mathbf{2 \text{ mark}}$$

$$0,128 = 2 \cdot 0,0325 + 2 \cdot n(H_3PO_4) \rightarrow n(H_3PO_4) = 0,0317 \text{ mmol} \quad \mathbf{2 \text{ mark}}$$

$$c_{M_1}(H_3PO_4) = \frac{n(H_3PO_4)}{V_{aliquota}(M_1)} = \frac{0,0317}{25,0} = 1,27 \cdot 10^{-3} \text{ M} \quad \mathbf{2 \text{ mark}}$$

Total = **12 mark**

16.3. Determine the phosphorus content (in mg) in the initial cereal sample. If you were unable to determine the phosphate concentration in the **M₁** solution, use $1,0 \cdot 10^{-3}$ mol/L for the calculations.

$$c_{M_0}(H_3PO_4) = \frac{c_{M_1}(H_3PO_4) \cdot V_{total}(M_1)}{V_{aliquota}(M_0)} = \frac{1,27 \cdot 10^{-3} \cdot 100}{25} = 5,08 \cdot 10^{-3} \text{ M} \quad \mathbf{2 \text{ mark}}$$

$$n_{M_0}(H_3PO_4) = c_{M_0}(H_3PO_4) \cdot V_{total}(M_0) = 5,08 \cdot 10^{-3} \cdot 250 = 1,27 \text{ mmol} \quad \mathbf{2 \text{ mark}}$$

$$n_{total}(P) = n_{M_0}(H_3PO_4) = 1,27 \text{ mmol} \cdot 31,0 \frac{\text{mg}}{\text{mmol}} = 39,4 \text{ mg} \quad \mathbf{2 \text{ mark}}$$

If the value $1,0 \cdot 10^{-3}$ mol/L is used, the final result is m(P) = 31,0 mg.

Total = **6 mark**

16.4. Calculate the pH of the M_1 solution obtained before performing the titration. If you were unable to determine the phosphate concentration in the M_1 solution, use $1.0 \cdot 10^{-3}$ mol/L for the calculations. Suppose sulfuric acid is strong in its two stages of dissociation.

Considering the completely dissociated sulfuric acid and only the first dissociation of phosphoric acid, the charge and mass balance equations for the system would be:

$$[H^+] = 2 \cdot [SO_4^{2-}] + [H_2PO_4^-]$$

$$[SO_4^{2-}] \approx c_{M_1}(H_2SO_4) = 1,30 \cdot 10^{-3} \text{ M}$$

$$[H^+] = [H_2PO_4^-] + 2 \cdot 1,30 \cdot 10^{-3} = [H_2PO_4^-] + 2,60 \cdot 10^{-3} \quad \text{2 mark}$$

$$c_{M_1}(H_3PO_4) = [H_3PO_4] + [H_2PO_4^-] = 1,27 \cdot 10^{-3}$$

$$[H_3PO_4] = 1,27 \cdot 10^{-3} - [H_2PO_4^-] \quad \text{2 mark}$$

$$K_{a1} = \frac{[H^+][H_2PO_4^-]}{[H_3PO_4]} \rightarrow 10^{-2,15} = \frac{([H_2PO_4^-] + 2,60 \cdot 10^{-3})[H_2PO_4^-]}{1,27 \cdot 10^{-3} - [H_2PO_4^-]} \quad \text{2 mark}$$

$$\text{Solving the quadratic equation, one gets } [H_2PO_4^-] = 5,09 \cdot 10^{-4} \text{ M} \quad \text{2 mark}$$

$$[H^+] = 8,58 \cdot 10^{-4} + 2 \cdot 1,27 \cdot 10^{-3} = 5,09 \cdot 10^{-4} + 2,60 \cdot 10^{-3} = 3,11 \cdot 10^{-3} \text{ M} \quad \text{2 mark}$$

$$\text{pH} = -\log(3,11 \cdot 10^{-3}) \approx 2,5 \quad \text{2 mark}$$

It is penalized with -2 mark if only the dissociation of H_2SO_4 is taken into account.

Total = **12 mark**

Part II. The above method may be compromised by the presence of other acidic species in the sample. A selective alternative for phosphate quantification is spectrophotometric determination via formation of the molybdo blue complex. In this assay, phosphate reacts in an acidic medium with ammonium molybdate to form a complex that, upon reduction, produces an intense blue coloration with a maximum absorption at 820 nm. The measurement of absorbed radiation (absorbance, A) is proportional to the concentration of the absorbing species, thereby allowing quantification.

A 10.0 mL aliquot of the M_0 stock solution was taken and transferred to a 100 mL volumetric flask. The reagents necessary for color development were added, and the solution was flushed with distilled water (M_2 Solution). The absorbance of a 25.0 mL aliquot of this dilute solution was measured, obtaining an A1 reading of 0.405. To a second aliquot of 25.0 mL of the same solution, 5.0 mL of a standard phosphate solution with a concentration of 0.200 mg/mL was added. The resulting mixture has an A2 absorbance of 0.667 under the same experimental conditions.

16.5. Calculate the phosphate concentration (in mg/mL) in the M_2 solution. Consider that the Lambert-Beer Law, $A = k \cdot c(X)$, is true, with k being constant for both measures.

$$A_1 = k \cdot c_{M_2}(PO_4^{3-}) = 0,405 \quad (\text{equation 1}) \quad \text{2 mark}$$

In the second experiment, a dilution occurred.

$$A_2 = k \cdot c'(PO_4^{3-}) = k \cdot \left[\frac{c_{M_2}(PO_4^{3-}) \cdot 25,0 + 0,200 \cdot 5,0}{25,0 + 5,0} \right] = 0,667 \quad (\text{equation 2}) \quad \text{4 mark}$$

Substituting the first equation into the second, we obtain

$$\frac{0,405}{c_{M_2}(PO_4^{3-})} \cdot \left[\frac{c_{M_2}(PO_4^{3-}) \cdot 25,0 + 0,200 \cdot 5,0}{25,0 + 5,0} \right] = 0,667$$

$$c_{M_2}(PO_4^{3-}) = 0,0410 \text{ mg/mL} \quad \text{4 mark}$$

Total = **10 mark**

16.6. Determine the phosphorus content (in mg) in the initial cereal sample. If you were unable to determine the phosphate concentration in the M_2 solution, use 0.0400 mg/mL in your calculations.

$$c_{M0}(PO_4^{3-}) = \frac{c_{M2}(PO_4^{3-}) \cdot V_{total}(M2)}{V_{aliquote}(M0)} = \frac{0,0410 \cdot 100}{10} = 0,410 \text{ mg/mL} \quad \mathbf{2 \text{ mark}}$$

$$m_{M0}(PO_4^{3-}) = c_{M0}(PO_4^{3-}) \cdot V_{total}(M0) = 0,410 \cdot 250 = 102,5 \text{ mg} \quad \mathbf{2 \text{ mark}}$$

$$\omega(P, PO_4^{3-}) = \frac{A_r(P)}{A_r(P) + 4 \cdot A_r(O)} = \frac{31,0}{31,0 + 4 \cdot 16,0} = 0,326 \quad \mathbf{2 \text{ mark}}$$

$$m_{total}(P) = 0,326 \cdot 102,5 \approx 33,1 \text{ mg} \quad \mathbf{2 \text{ mark}}$$

Total = **8 mark**Considering the value of $c_{M2}(PO_4^{3-}) = 0,0400 \text{ mg/mL}$ it is obtained that $m_{total}(P) \approx 32,3 \text{ mg}$

Part III. As a third method of quantification, phosphate was determined by quantitative precipitation as oxovanadium(IV) phosphate, $(VO)_3(PO_4)_2$. A 100 mL aliquot of the M_0 solution was taken, slightly acidified and heated to 60 °C. Subsequently, an excess of 0.050 M standard $VOSO_4$ solution was slowly added to ensure complete precipitation. After filtration, washing and drying in the oven to constant mass, a precipitate mass reading of 80.4 mg was obtained.

16.7. Determine the phosphorus content (in mg) in the initial cereal sample.

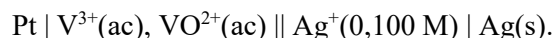
$$\omega(P, pp) = \frac{2 \cdot A_r(P)}{3 \cdot A_r(V) + 2 \cdot A_r(P) + 11 \cdot A_r(O)} = \frac{2 \cdot 31,0}{3 \cdot 50,9 + 2 \cdot 31,0 + 11 \cdot 16,0} = 0,159 \quad \mathbf{2 \text{ mark}}$$

$$m_{pp}(P) = 0,159 \cdot 80,4 \approx 12,8 \text{ mg} \quad \mathbf{2 \text{ mark}}$$

$$m_{total}(P) = \frac{250}{100} \cdot 12,8 = 32,0 \text{ mg} \quad \mathbf{2 \text{ mark}}$$

Total = **6 mark**

Part IV. To determine the solubility product (Kps) and molar solubility of the $(VO)_3(PO_4)_2$, a sufficient amount of the solid was brought into contact with a buffering solution of pH 6.0 acetate. A controlled amount of V^{3+} was added to this system, and the system was kept in agitation until equilibrium was reached. By ICP-OES analysis, the total concentration of vanadium ($V^{3+} + VO^{2+}$) dissolved in the supernatant was determined to be $1,0 \cdot 10^{-4} \text{ M}$. The following electrochemical cell was then constructed using the saturated solution of vanadium ions as the anodic electrolyte:



The measured cell potential was 1.128 V at 25 °C. $E^\circ(VO^{2+}/V^{3+}) = 0.337 \text{ V}$, $E^\circ(Ag^+/Ag) = 0.799 \text{ V}$.

16.8. Calculate the solubility product constant of $(VO)_3(PO_4)_2$.

The silver electrode is not at standard conditions, so the potential must be corrected.

$$E(Ag^+, Ag) = E^\circ - 0,0591 \cdot \log \frac{1}{[Ag^+]} = 0,799 - 0,0591 \cdot \log \left(\frac{1}{0,100} \right) = 0,740 \text{ V} \quad \mathbf{2 \text{ mark}}$$

$$E(VO^{2+}, V^{3+}) = E(Ag^+, Ag) - \Delta E = 0,740 - 1,128 = -0,388 \text{ V} \quad \mathbf{2 \text{ mark}}$$

The vanadium electrode is also not in standard condition.



$$E(VO^{2+}, V^{3+}) = E^\circ - 0,0591 \cdot \log \frac{[V^{3+}]}{[VO^{2+}][H^+]^2} \quad \mathbf{2 \text{ mark}}$$

$$-0,388 = 0,337 - 0,0591 \cdot \log \frac{[V^{3+}]}{[VO^{2+}](10^{-6})^2} \rightarrow \frac{[V^{3+}]}{[VO^{2+}]} = 1,849 \quad \mathbf{2 \text{ mark}}$$

$$[VO^{2+}] + [V^{3+}] = 1,0 \cdot 10^{-4} \text{ M} \rightarrow [V^{3+}] = 6,49 \cdot 10^{-5} \text{ M} \quad \mathbf{0 \text{ mark}} \quad [VO^{2+}] = 3,51 \cdot 10^{-5} \text{ M} \quad \mathbf{2 \text{ mark}}$$

$$c_{total}(PO_4^{3-}) = \frac{2}{3} \cdot [VO^{2+}] = \frac{2}{3} \cdot 3,51 \cdot 10^{-5} = 2,34 \cdot 10^{-5} \text{ M} \quad \mathbf{2 \text{ mark}}$$

$$\alpha_3 = \frac{K_{a1}K_{a2}K_{a3}}{[H^+]^3 + K_{a1}[H^+]^2 + K_{a1}K_{a2}[H^+] + K_{a1}K_{a2}K_{a3}} \quad \mathbf{2 \text{ mark}}$$

$$\alpha_3 = \frac{10^{-2,15} \cdot 10^{-7,20} \cdot 10^{-12,32}}{[10^{-6}]^3 + 10^{-2,15} \cdot [10^{-6}]^2 + 10^{-2,15} \cdot 10^{-7,20} \cdot 10^{-6} + 10^{-2,15} \cdot 10^{-7,20} \cdot 10^{-12,32}} = 2,84 \cdot 10^{-8} \quad \text{2 mark}$$

$$K_{ps} = [VO^{2+}]^3 [PO_4^{3-}]^2 = [VO^{2+}]^3 [\alpha_3 \cdot c_{total}(PO_4^{3-})]^2 \quad \text{2 mark}$$

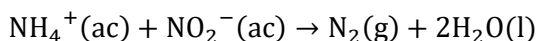
$$K_{ps} = [3,51 \cdot 10^{-5}]^3 [2,84 \cdot 10^{-8} \cdot 2,34 \cdot 10^{-5}]^2 = 1,9 \cdot 10^{-38} \quad \text{2 mark}$$

Total = 22 mark

Question 17. Kinetics of ammonium cation (20% of total; only 12th grade) **Day 2**

17.1	17.2	17.3	17.4	17.5	17.6	17.7	17.8	17.9	17.10	17.11	17.12	17.13	17.14	17.15	Total	Total
5	4	8	8	3	12	3	4	4	3	2	2	14	4	4	80	20
5	4	8	8	3	12	3	4	4	3	2	2	14	4	4	80	20

The youngest student of the last Preselection is called Cristhian, and he is passionate about Inorganic and Physical Chemistry. In the laboratory of the Faculty of Chemistry, he designed an experiment to determine the rate law for the reaction between nitrite and ammonium ions.



Cristhian took ten clean, dry containers. In five labeled containers, he added different volumes of ammonium chloride solution (5.0 mol/L); in the other five equally labeled containers, he added different volumes of sodium nitrite solution (5.0 mol/L), as indicated in the table. Then, he mixed the ammonium chloride solutions with the sodium nitrite solutions in an orderly manner, added distilled water and an inert phosphate buffer of pH = 3 to obtain a total volume of 100 mL of solution, although with different concentrations, and weighed each mixture. After 2 minutes of reaction, he weighed the mixtures again.

The difference in mass Δm for each experiment, the product of the reaction that occurred, is shown in the following table.

Exp.	Δm (g)	T (°C)	$V_0(\text{NH}_4^+)$ (mL)	$V_0(\text{NO}_2^-)$ (mL)	$V_0(\text{H}_2\text{O})$ (mL)	$V_0(\text{buffer})$ (mL)
1	0,151	25	20	20	50	10
2	0,236	25	20	25	45	10
3	0,085	25	20	15	55	10
4	0,106	25	25	15	50	10
5	0,064	25	15	15	60	10

17.1. Prove that the above reaction occurs almost completely by calculating the equilibrium thermodynamic constant. E° (V): $\text{N}_2/\text{NH}_4^+ = -0.28$; $\text{NO}_2^-/\text{N}_2 = 1.45$.

$$\Delta E^\circ = E^\circ(\text{NO}_2^-, \text{N}_2) - E^\circ(\text{N}_2, \text{NH}_4^+) = 1,45 - (-0,28) = 1,73 \text{ V} \quad \text{2 mark}$$

$$K^\circ = \exp\left(\frac{zF\Delta E^\circ}{RT}\right) = \exp\left(\frac{3 \cdot 2,303 \cdot 1,73}{0,0591}\right) = 6,8 \cdot 10^{87} \quad \text{3 mark}$$

Total = **5 mark**

17.2. Explain why: a) the NH_4^+ cation is a Lewis acid, b) the NO_2^- anion is an ambidentate base, c) the dinitrogen is very stable, and d) the water is amphoteric.

a) NH_4^+ is a Lewis acid because it can accept electrons through its positive hydrogen.

b) Nitrite is an ambivalent base because it can donate the electron pairs of N and O atoms.

c) The high stability of dinitrogen is because it has a triple bond between its atoms.

d) The water is self-ionized to form the ions H^+ (acid) and OH^- (base).

Total = **4 mark**

17.3. With the data obtained, Cristhian calculated the reaction rate for each experiment, expressed as moles of N_2 formed per liter of solution per second. What values did he obtain?

$$v_r = \frac{d[\text{N}_2]}{dt} = \frac{1}{V_D} \frac{dn(\text{N}_2)}{dt} \approx \frac{\Delta m(\text{N}_2)}{V_D \cdot M(\text{N}_2) \cdot \Delta t} \quad \text{3 mark}$$

$$\text{Exp 1} \quad v_r \approx \frac{0,151}{0,1 \cdot 28,0 \cdot 2 \cdot 60} = 4,49 \cdot 10^{-4} \text{ Ms}^{-1} \quad \text{1 mark}$$

$$\text{Exp 2} \quad v_r \approx \frac{0,236}{0,1 \cdot 28,0 \cdot 2 \cdot 60} = 7,02 \cdot 10^{-4} \text{ Ms}^{-1} \quad \mathbf{1 \text{ mark}}$$

$$\text{Exp 3} \quad v_r \approx \frac{0,085}{0,1 \cdot 28,0 \cdot 2 \cdot 60} = 2,53 \cdot 10^{-4} \text{ Ms}^{-1} \quad \mathbf{1 \text{ mark}}$$

$$\text{Exp 4} \quad v_r \approx \frac{0,106}{0,1 \cdot 28,0 \cdot 2 \cdot 60} = 3,15 \cdot 10^{-4} \text{ Ms}^{-1} \quad \mathbf{1 \text{ mark}}$$

$$\text{Exp 5} \quad v_r \approx \frac{0,064}{0,1 \cdot 28,0 \cdot 2 \cdot 60} = 1,90 \cdot 10^{-4} \text{ Ms}^{-1} \quad \mathbf{1 \text{ mark}}$$

Total = **8 mark**

17.4. What is the expression for the rate law, rounding orders to integers?

Let's build a new table with the concentration values of the reactants:

Exp.	$[\text{NH}_4^+]_0$ (M)	$[\text{NO}_2^-]_0$ (M)	v_0 (Ms^{-1})	$\ln [\text{NH}_4^+]_0$	$\ln [\text{NO}_2^-]_0$	$\ln v_0$
1	1,0	1,0	$4,49 \cdot 10^{-4}$	0,00	0,00	-7,71
2	1,0	1,25	$7,02 \cdot 10^{-4}$	0,00	0,223	-7,26
3	1,0	0,75	$2,53 \cdot 10^{-4}$	0,00	-0,288	-8,28
4	1,25	0,75	$3,16 \cdot 10^{-4}$	0,223	-0,288	-8,06
5	0,75	0,75	$1,89 \cdot 10^{-4}$	-0,288	-0,288	-8,57

$$\text{Let } v_r = k_r \cdot [\text{NH}_4^+]^a \cdot [\text{NO}_2^-]^b$$

The data from the first three experiments (at $[\text{NH}_4^+]_0 = 1 \text{ M}$, constant) adjust to the straight line

$$\ln v_0 = 2,0 \cdot [\text{NO}_2^-] - 7,71 \quad (r = 1). \text{ So the order of reaction } b = 2. \quad \mathbf{3 \text{ mark}}$$

Data from the last three experiments (at $[\text{NO}_2^-]_0 = 0,75 \text{ M}$, constant) adjust to the straight line

$$\ln v_0 = 1,0 \cdot [\text{NH}_4^+] - 8,28 \quad (r = 1). \text{ So the order of reaction } a = 1. \quad \mathbf{3 \text{ mark}}$$

$$\text{Rate Law } v_r = k_r \cdot [\text{NH}_4^+] \cdot [\text{NO}_2^-]^2 \quad \mathbf{2 \text{ mark}}$$

No points are awarded for calculating the concentrations of the reactants, as it is not essential to solve the problem. Other solutions that do not involve linear regressions are also accepted. Total = **8 mark**

17.5. Calculate the apparent rate constant of the reaction at 25 °C.

$$k_{\text{exp}} = \frac{v_r}{[\text{NH}_4^+] \cdot [\text{NO}_2^-]^2} = \frac{4,49 \cdot 10^{-4}}{1,0 \cdot 1,0^2} = 4,49 \cdot 10^{-4} \text{ M}^{-2} \text{ s}^{-1} \quad \mathbf{3 \text{ mark}}$$

Total = **3 mark**

17.6. Help Cristhian to determine the activation energy (in kJ/mol) of the reaction, knowing the values of the specific rate constant for each temperature. To do this, perform a linear regression (provide the value of r) or make a graph, using the Arrhenius equation.

T (°C)	25	30	35	40	45	50
k_r (u)	0,45	0,68	1,0	1,5	2,2	3,2

The linearized Arrhenius equation is $\ln k_r = -\frac{E_A}{R} \cdot \left(\frac{1}{T}\right) + \ln A$ **6 mark** by data table.

$1/T$ (K^{-1})	$3,35 \cdot 10^{-3}$	$3,30 \cdot 10^{-3}$	$3,25 \cdot 10^{-3}$	$3,19 \cdot 10^{-3}$	$3,14 \cdot 10^{-3}$	$3,09 \cdot 10^{-3}$
$\ln k_r$	-0,80	-0,39	0,00	0,41	0,79	1,16

The data fit the equation. $\ln k_r = -7469 \cdot (1/T) + 24,2$ ($r = 1$) **4 mark**

$$E_A = 7469 \cdot 8,314 \cdot 10^{-3} \approx 62 \text{ kJ/mol} \quad \mathbf{2 \text{ mark}}$$

Total = **12 mark**

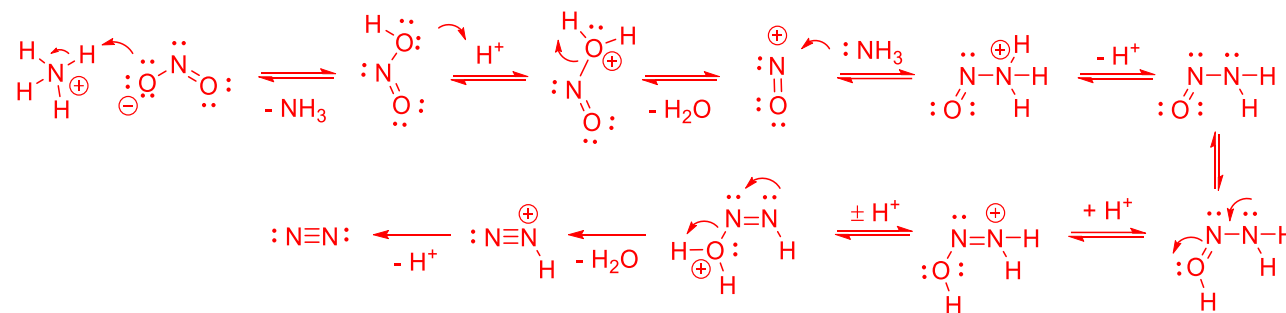
17.7. It is known that the rate of this reaction depends on the concentration of protons. Using the data in the table above and the constant you calculated in **17.3**, determine the order of reaction with respect to $[H^+]$.

$$k_{\text{exp}} = k_r \cdot [H^+]^c \rightarrow 4,49 \cdot 10^{-4} = 0,45 \cdot [1,0 \cdot 10^{-3}]^c \rightarrow c = 1$$

Total = **3 mark**

17.8. When Cristhian determined the total order, he realized that the reaction had to occur by a complex mechanism, which can be ionic or radical. Propose a mechanism showing the flow of electrons using arrows. Consider that H_2NNO is one of the intermediates that is formed.

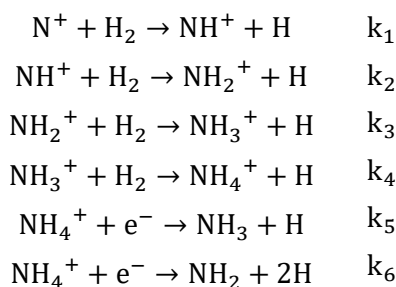
A possible mechanism via the ionic pathway may be as follows.



Other logical mechanisms are accepted, including the radical pathway.

Total = **4 mark**

Adolfo is the tallest student in the pre-selection and is very curious. After seeing the experiment above, he wondered if that reaction could happen in space. While searching the literature for data, he found that at least the ammonium cation is present in interstellar gas clouds and is involved in the following complex mechanism.



17.9. What are the boundary orbitals involved in the reaction of the first step of the above mechanism? Classify them into HOMO, LOMO, or SOMO.

For the N^+ reactant, the OF is the 2p atomic orbital of the SOMO type **2 mark**

For the H_2 reactant, the OF is the molecular orbital σ^* of the LUMO type **2 mark**

Total = **4 mark**

17.10. The ionization energy of the nitrogen atom is 14.53 eV. Help Adolf calculate the maximum wavelength a photon can have so that it is capable of forming the N^+ ion.

$$E = h \cdot \frac{c}{\lambda} \rightarrow 14,53 \cdot 1,602 \cdot 10^{-19} = 6,626 \cdot 10^{-34} \cdot \frac{2,998 \cdot 10^8}{\lambda}$$

$$\lambda = 8,534 \cdot 10^{-8} \text{ m} = 85,3 \text{ nm in the extreme ultraviolet of the electromagnetic spectrum Total = 3 mark}$$

17.11. In reality, the NO_2^- anion is not stable in space. Why?

In space, high radiant energy causes nitrite to rapidly lose an electron (become ionized), transforming it into a neutral oxide and other products.

Total = **2 mark**

17.12. Sort the N^+ , NH^+ , NH_2^+ , NH_3^+ , and NH_4^+ species with respect to their electron affinity.

The electron affinity will be higher in the system with less dispersion of the positive charge and a higher average electronegativity of its atoms. $N^+ > NH^+ > NH_2^+ > NH_3^+ > NH_4^+$ Total = **2 mark**

17.13. Use the steady-state approximation to derive equations for the concentrations of the intermediates NH^+ , NH_2^+ , NH_3^+ , and NH_4^+ in terms of the concentrations of $[N^+]$, $[H_2]$, and $[e^-]$ only.

$$\frac{d[NH^+]}{dt} = 0 = k_1[N^+][H_2] - k_2[NH^+][H_2] \rightarrow [NH^+] = \frac{k_1[N^+]}{k_2} \quad \text{2 mark}$$

$$\frac{d[NH_2^+]}{dt} = 0 = k_2[NH^+][H_2] - k_3[NH_2^+][H_2] \rightarrow [NH_2^+] = \frac{k_2[N^+]}{k_3} \quad \text{2 mark}$$

$$[NH_2^+] = \frac{k_2[NH^+]}{k_3} = \frac{k_2}{k_3} \cdot \frac{k_1[N^+]}{k_2} = \frac{k_1[N^+]}{k_3} \quad \text{2 mark}$$

$$\frac{d[NH_3^+]}{dt} = 0 = k_3[NH_2^+][H_2] - k_4[NH_3^+][H_2] \rightarrow [NH_3^+] = \frac{k_3[N^+]}{k_4} \quad \text{2 mark}$$

$$[NH_3^+] = \frac{k_3[NH_2^+]}{k_4} = \frac{k_3}{k_4} \cdot \frac{k_1[N^+]}{k_3} = \frac{k_1[N^+]}{k_4} \quad \text{2 mark}$$

$$\frac{d[NH_4^+]}{dt} = 0 = k_4[NH_3^+][H_2] - k_5[NH_4^+][e^-] - k_6[NH_4^+][e^-] \rightarrow [NH_4^+] = \frac{k_4[NH_3^+][H_2]}{(k_5+k_6)[e^-]} \quad \text{2 mark}$$

$$[NH_4^+] = \frac{k_4[NH_3^+][H_2]}{(k_5+k_6)[e^-]} = \frac{k_4[H_2]}{(k_5+k_6)[e^-]} \cdot \frac{k_1[N^+]}{k_4} = \frac{k_1[N^+][H_2]}{(k_5+k_6)[e^-]} \quad \text{2 mark} \quad \text{Total} = \text{14 mark}$$

17.14. Demonstrate that the rate of ammonia formation depends only on the concentration of N^+ and H_2 . Find an expression for k in terms of the velocity constants of the elementary steps.

$$\frac{d[NH_3]}{dt} = k \cdot [N^+][H_2]$$

$$\frac{d[NH_3]}{dt} = k_5[NH_4^+][e^-] = k_5[e^-] \frac{k_1[N^+][H_2]}{(k_5+k_6)[e^-]} = \frac{k_1 k_5 [N^+][H_2]}{(k_5+k_6)} = k \cdot [N^+][H_2] \quad \text{2 mark}$$

$$k = \frac{k_1 k_5}{(k_5+k_6)} \quad \text{2 mark}$$

Total = **4 mark**

In his research, Adolfo found that the rates of many reactions involving ions and simple molecules with radicals show virtually no temperature dependence.

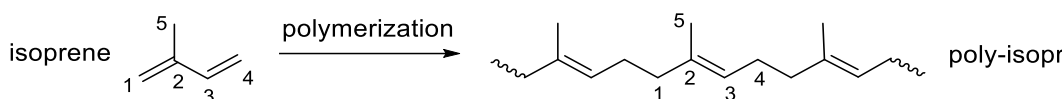
17.15. Considering the origin of the activation energy, explain the above fact and discuss its relevance to the reactions that occur in interstellar space.

The reactions described have practically zero activation energy. This occurs because there is no need to break strong bonds or rearrange the molecular geometry. In interstellar space, temperatures are very low, so only these reactions can occur at an appreciable rate. Total = **4 mark**

Question 18. Luffy's skin (20% of total; only 12th grade) Day 2

18.1	18.2	18.3	18.4	18.5	18.6	18.7	18.8	Total	Total
8	18	6	14	10	6	10	8	80	20
8	18	6	14	10	6	10	8	80	20

Luffy is the main character of the Japanese anime series One Piece and is very popular among students for his bravery, great strength, sense of justice, and free spirit. Luffy's skin is made of polyisoprene, the main component that gives natural rubber its elasticity.



18.1. Which of the following reagents are capable of inducing isoprene polymerization?

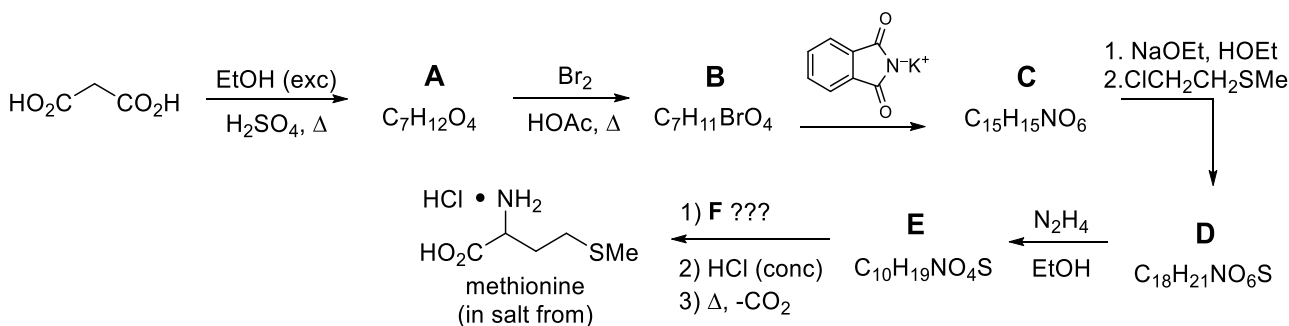
Reagent: a) $\text{K}_2\text{S}_2\text{O}_8$; b) H_2O ; c) BuLi ; d) BF_3 ; e) NaIO_4 ; f) $(\text{PhCO}_2)_2$; g) Zn , HCl ; h) $\text{Na}(\text{Hg})$; i) Br_2 ; j) I_2 , NaOH ; k) EtONa ; l) TiCl_4 ; m) O_3 ; n) EtOLi , Et_2Zn ; o) H_2 , Pd ; p) NaBH_4 ; q) LDA

a) $\text{K}_2\text{S}_2\text{O}_8$; c) BuLi ; d) BF_3 ; f) $(\text{PhCO}_2)_2$; h) $\text{Na}(\text{Hg})$; l) TiCl_4 ; n) EtOLi , Et_2Zn ; q) LDA

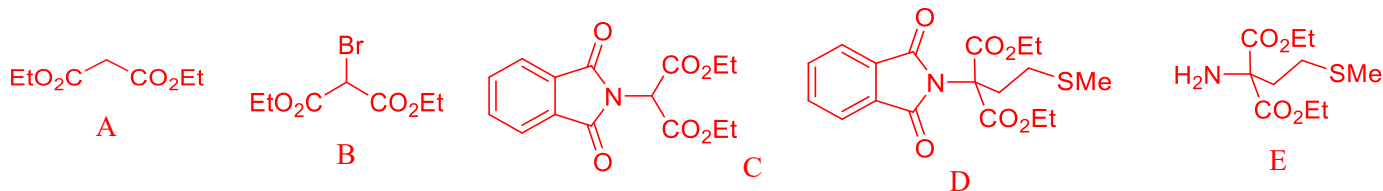
–1 mark is penalized for incorrect reagents.

Total = **8 mark**

Luffy needs to consume moderate amounts of sulfur; so their diet should be rich in the natural amino acids cysteine and methionine, which are very abundant in foods such as meats, eggs, nuts and legumes. When food is scarce during long voyages in the Blue Sea, the crew doctor, Dr. Chopper, who is also an excellent chemist, prepares a food supplement of artificial methionine in the laboratory of the Thousand Sunny by synthesis with potassium phthalimide:



18.2. Represent the structures of the A-E compounds and suggest the F reagent.



F = NaOH/KOH/LiOH **3 mark** for each answer

Total = **18 mark**

Luffy's metabolism is peculiar. Methionine and cysteine are enzymatically transformed into elemental sulfur, which in turn reacts with the poly-isoprene present in the skin through a process known as "vulcanization", forming C-S-C, C-S-S-C and C-S_n-C bonds between the polymer chains, and giving it greater strength and elasticity to the skin.

18.3. Name which positions (1, 2, 3, 4, or 5) of the polymer are most susceptible to the vulcanization reaction, and order the three types of bonds formed during vulcanization according to their thermal stability.

Allylic positions 1, 4 and 5 **3 mark** C-S-C > C-S-S-C > C-S_n-C **3 mark** Total = **6 mark**

However, excess sulfur can be harmful, causing Luffy's skin to become hard and brittle. In these cases, Chopper prepares a special cream containing 85% sodium stearate CH₃(CH₂)₁₆CO₂Na, an excipient or additive) and 15% Ni-Raney (an active component). The salt is obtained by saponification of natural triglycerides. Ni-Raney is obtained by reaction of an equimolar alloy (1:1) of nickel and aluminum with concentrated and hot sodium hydroxide, until all the aluminum is consumed and only chemisorbed hydrogen remains onto finely divided nickel (of NiH_x composition, 1 < x < 2).

18.4. What minimum masses of nickel, aluminum, sodium hydroxide and stearate are needed to prepare one kilogram of special Chopper cream? Suppose that the Ni-Raney has NiH_{1.5} composition and that the aluminum in the alloy is dissolved in the form of [Al(OH)₄]⁻.

$$m(\text{sal}) = \frac{85}{100} \cdot 1000 = 850 \text{ g} \quad \mathbf{2 \text{ mark}}$$

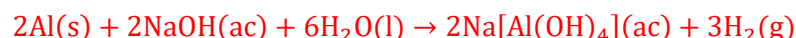
$$m(\text{NiH}_{1.5}) = \frac{15}{100} \cdot 1000 = 150 \text{ g} \quad \mathbf{2 \text{ mark}}$$

$$M(\text{NiH}_{1.5}) = 58,7 + 1,5 \cdot 1,0 = 60,2 \text{ g} \cdot \text{mol}^{-1} \quad n(\text{NiH}_{1.5}) = \frac{150}{60,2} = 2,49 \text{ mol}$$

$$n(\text{Ni}) = n(\text{NiH}_{1.5}) = 2,49 \text{ mol} \cdot 58,7 \frac{\text{g}}{\text{mol}} = 146 \text{ g} \quad \mathbf{3 \text{ mark}}$$

$$n(\text{Al}) = n(\text{Ni}) = 2,49 \text{ mol} \cdot 27,0 \frac{\text{g}}{\text{mol}} = 67,2 \text{ g} \quad \mathbf{3 \text{ mark}}$$

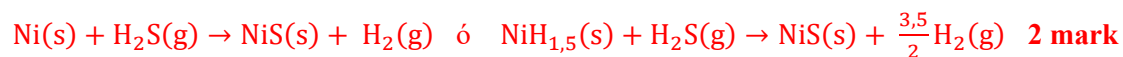
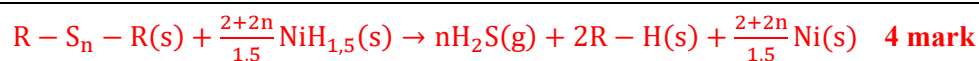
Depending on the conditions of the problem, not all of the dihydrogen produced becomes part of the hydride. So the limiting substance is aluminum.



The counterion of the complex anion must be sodium, from which it follows that.

$$n(\text{NaOH}) = n(\text{Al}) = 2,49 \text{ mol} \cdot (23,0 + 16,0 + 1,0) \frac{\text{g}}{\text{mol}} = 99,6 \text{ g} \quad \mathbf{4 \text{ mark}} \quad \text{Total} = \mathbf{14 \text{ mark}}$$

18.5. During the skin treatment with special cream, sailor Nami and cook Sanji are always in a bad mood, because the whole boat takes on a very unpleasant smell. Formulate the equations for the chemical reactions that occur, given that a black solid is one of the products. Suppose the polymer formula of Luffy's skin is R-S_n-R when fully vulcanized.

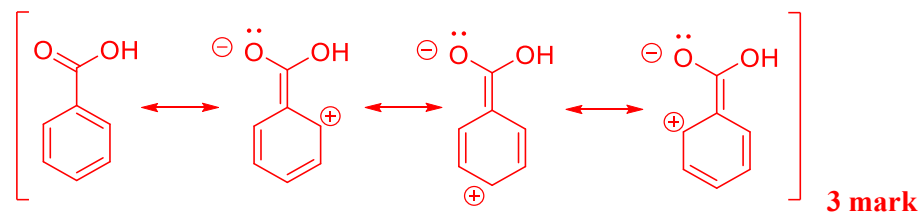
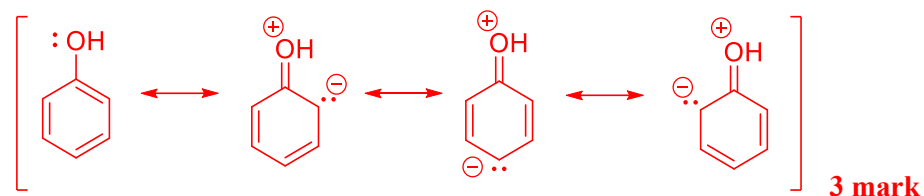


Other similar equations, such as the direct formation NiS from the polymer, are accepted as long as they are logical and balanced. Total = **6 mark**

During the assault on Impel Down Prison to rescue his brother, Luffy was attacked by the head manager, Magellan, whose body is covered in highly toxic substances such as Br₂, H₂SO₄ and HNO₃, concentrated and hot. Fortunately, Luffy's immune system produces antidotes for these poisons, which facilitate their elimination, for example, p-hydroxybenzoic acid.

18.6. Classify the hydroxyl and carboxylic acid groups according to their electronic effect on the S_EAr reaction. Justify your answer by the resonant structures of phenol and benzoic acid.

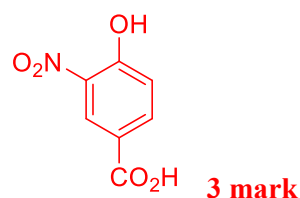
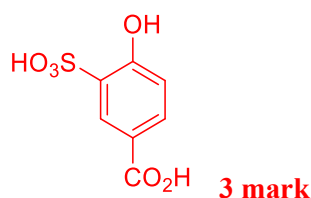
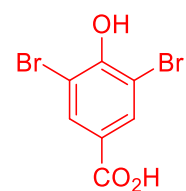
hydroxyl = activator or electrodonor **2 mark** acid = deactivator or electroacceptor **2 mark**



Only resonant structures with the charge delocalized by the ring will be scored.

Total = **10 mark**

18.7. Represent the structures of the main reaction products of p-hydroxybenzoic acid with Br_2 , H_2SO_4 and HNO_3 , separately. In the reaction with a halogen, a double substitution occurs.



Total = **10 mark**

18.8. What are the true electrophiles in S_EAr reactions with H_2SO_4 and HNO_3 ? Write the chemical equation for the electrophile formation reaction in each case.

SO_3 **2 mark** $H_2SO_4(ac) \rightarrow H_2O(l) + SO_3(ac)$ **2 mark**

NO_2^+ **2 mark** $HNO_3(ac) + H^+(ac) \rightarrow H_2O(l) + NO_2^+(ac)$ **2 mark**

Other electrophile formulations ($H_3SO_4^+$, SO_3H^+) and logical equations are accepted.

Total = **8 mark**